WEB-BASED SALES, INVENTORY, CUSTOMER, AND LOCATION MAPPING MANAGEMENT SYSTEM FOR A WATER REFILLING STATION FOR AQUA FABZ WATER REFILLING STATION

A Capstone Project

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BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

by

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CHAPTER 3

METHODOLOGY

Use Case

Figure 1.

Web-Based Sales, Inventory, Customer, and Location Mapping Management System Main Use Case Diagram

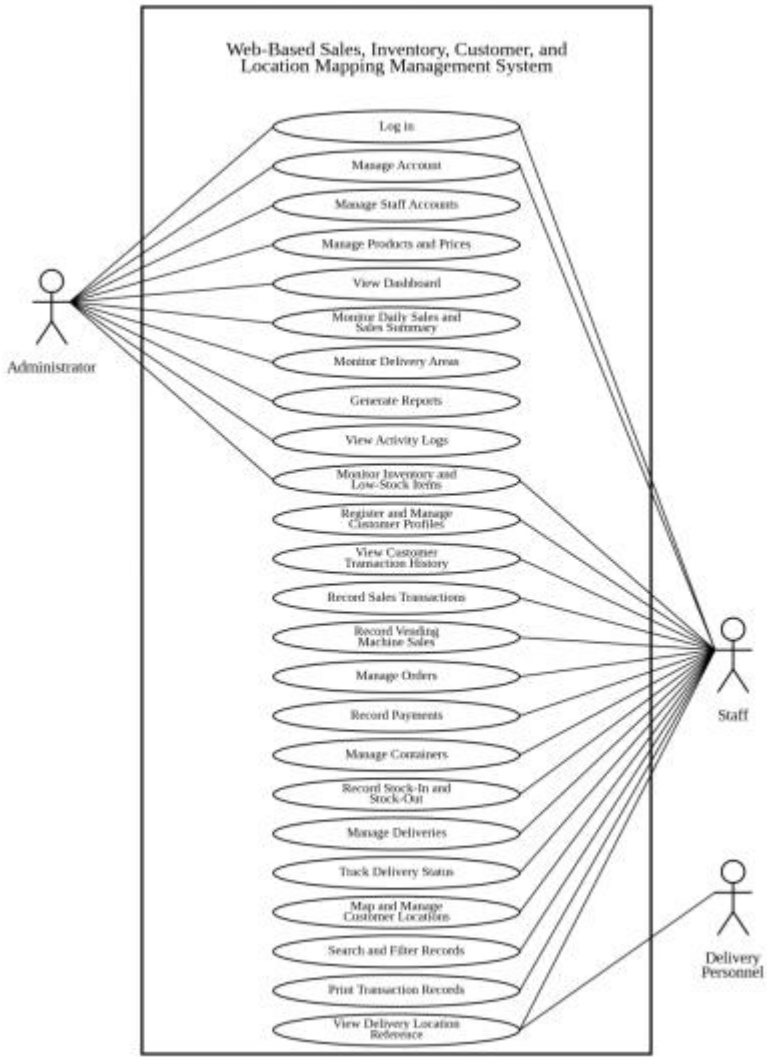


Figure 1 shows the main use case diagram of the system, illustrating the three actors (Administrator, Staff, and Delivery Personnel) and the 24 use cases within the system boundary.

Table 1 lists the 24 use cases counted from the diagram, in the order they appear from top to bottom, together with the actors connected to each use case. Delivery Personnel do not log in; they use the schedule, the delivery statuses, and the map reference that Staff provide.

Table 1:

Summary of Use Cases and Actors

No.	Use Case	Actor(s)
1	Log in	Administrator, Staff
2	Manage Account	Administrator, Staff
3	Manage Staff Accounts	Administrator
4	Manage Products and Prices	Administrator
5	View Dashboard	Administrator
6	Monitor Daily Sales and Sales Summary	Administrator
7	Monitor Delivery Areas	Administrator
8	Generate Reports	Administrator
9	View Activity Logs	Administrator
10	Monitor Inventory and Low-Stock Items	Administrator, Staff
11	Register and Manage Customer Profiles	Staff
12	View Customer Transaction History	Staff
13	Record Sales Transactions	Staff
14	Record Vending Machine Sales	Staff
15	Manage Orders	Staff
16	Record Payments	Staff
17	Manage Containers	Staff
18	Record Stock-In and Stock-Out	Staff
19	Manage Deliveries	Staff
20	Track Delivery Status	Staff
21	Map and Manage Customer Locations	Staff

22	Search and Filter Records	Staff
23	Print Transaction Records	Staff
24	View Delivery Location Reference	Staff, Delivery Personnel

Figure 2.

Log in Use Case Diagram

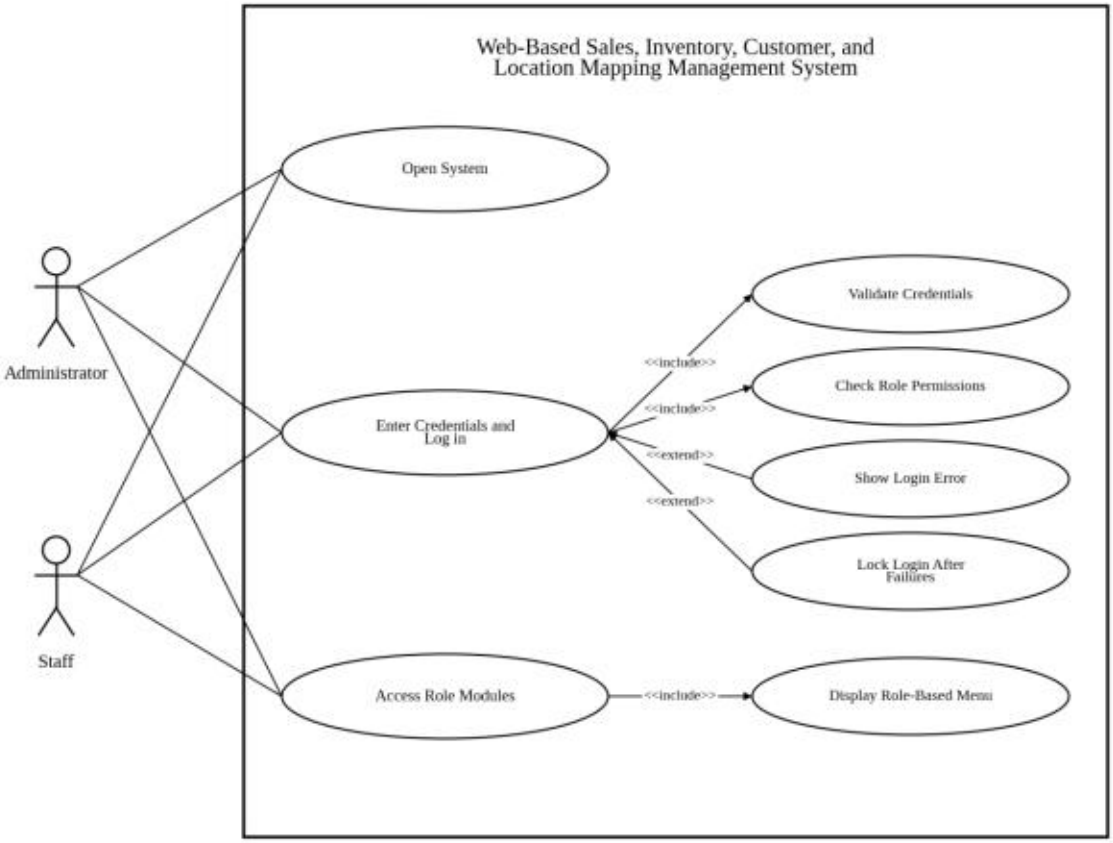


Figure 2 shows the use case diagram of Log in, which is performed by: Administrator, Staff. Each step of the Normal Flow in Table 2 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 2:

Log in Use Case Description

Use Case Title:	Log in
Description:	Allows the Administrator and Staff to log in securely and access only the functions of their assigned role and permissions.
Actors:	Administrator, Staff
Pre-Condition(s):	1. The system is online. 2. The user has a registered and active account.
Normal Flow:	

Actor	**System**
1. Open the system	1.1 Display the login page
2. Enter the username and password and click the Login button	2.1 Validate the credentials against the registered accounts <<include>> Validate Credentials 2.2 Identify the role and display only the modules allowed for that role <<include>> Check Role Permissions 2.3 <<extend>> Show Login Error 2.4 <<extend>> Lock Login After Failures
3. Access the modules of his or her role	3.1 Display the home page and the menu of the user's role <<include>> Display Role-Based Menu

Alternative Flow:	**A1. Invalid credentials** A1.1 The system displays an error message and denies access. A1.2 After repeated failed attempts, the system temporarily locks the login. **A2. Staff user opens an Administrator-only page** A2.1 The system denies access and displays a message. **A3. Account is deactivated** A3.1 The system denies access and informs the user to contact the Administrator.
Post-Condition(s):	1. The user is logged in and directed to the modules of his or her role. 2. If login fails, the user remains logged out.

Business Rules:	1. Only active accounts can log in. 2. Each account is assigned one role only: Administrator or Staff. 3. Accounts are created by the Administrator through Manage Staff Accounts; there is no self-registration. 4. Delivery Personnel and customers do not have login accounts.

Figure 3.

Manage Account Use Case Description

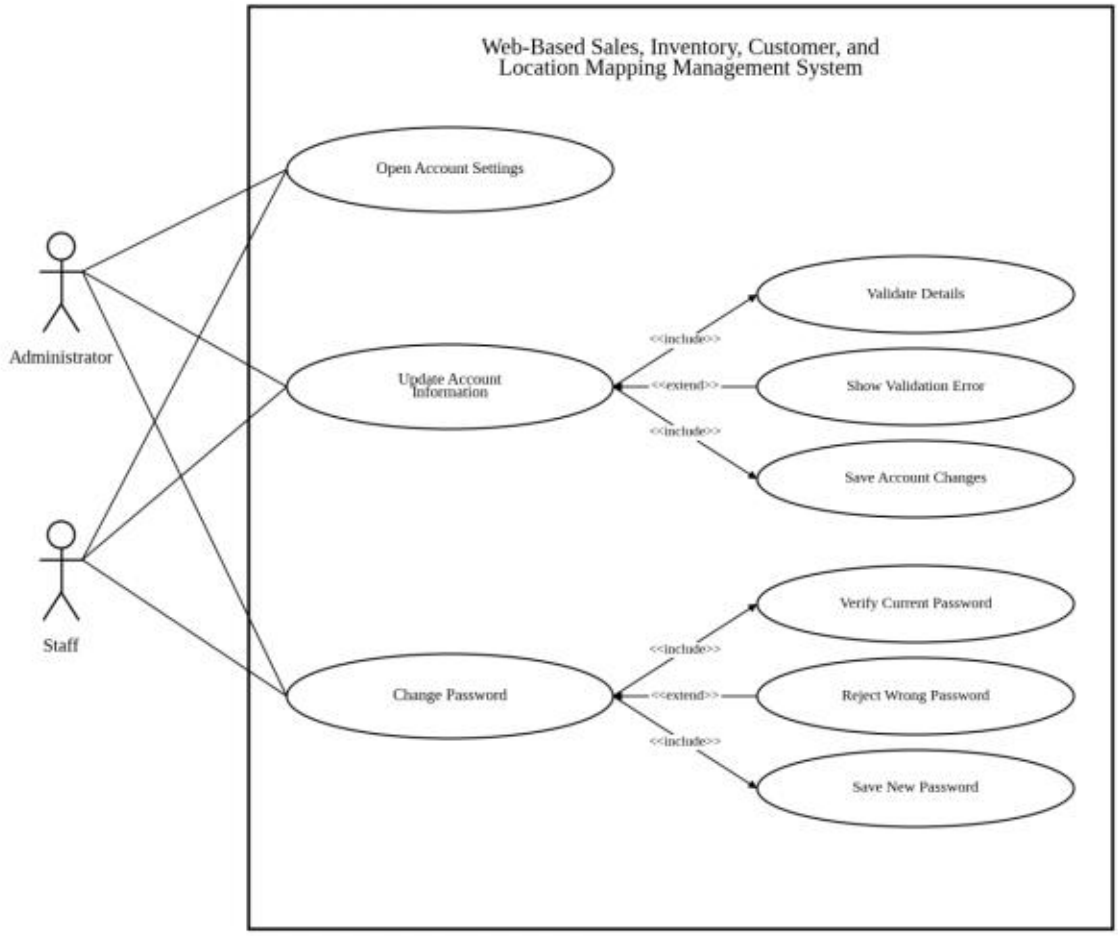


Figure 3 shows the use case diagram of Manage Account, which is performed by: Administrator, Staff. Each step of the Normal Flow in Table 3 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 3:

Manage Account Use Case Description

<table>
<tr><td>Use Case Title:</td><td>Manage Account</td></tr>
<tr><td>Description:</td><td>Allows the Administrator and Staff to update their own account information and password.</td></tr>
<tr><td>Actors:</td><td>Administrator, Staff</td></tr>
<tr><td>Pre-Condition(s):</td><td>The user must be logged in.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open Account Settings</td><td>1.1 Display the current account details of the logged-in user</td></tr>
<tr><td>2. Update the account information and save</td><td>2.1 Validate the entered details <<include>> Validate Details
2.2 <<extend>> Show Validation Error
2.3 Save the changes and display a confirmation message <<include>> Save Account Changes</td></tr>
<tr><td>3. Enter the current password and the new password and save</td><td>3.1 Verify the current password <<include>> Verify Current Password
3.2 <<extend>> Reject Wrong Password
3.3 Save the new password and record the change in the activity log <<include>> Save New Password</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Required fields are incomplete or invalid
A1.1 The system displays a validation error message and does not save the changes.
A1.2 The user corrects the entries and saves again.
A2. Current password is incorrect
A2.1 The system rejects the change and informs the user.
A2.2 The user enters the correct current password.
A3. New password and confirmation do not match
A3.1 The system asks the user to enter the same new password in both fields.</td></tr>
</table>

Post-Condition(s):	1. The updated account information or password is saved. 2. A password change is recorded in the activity log.
Business Rules:	1. Each user can manage only his or her own account. 2. A user cannot change his or her own role. 3. The current password is required to set a new password.

Figure 4.

Manage Staff Accounts Use Case Diagram

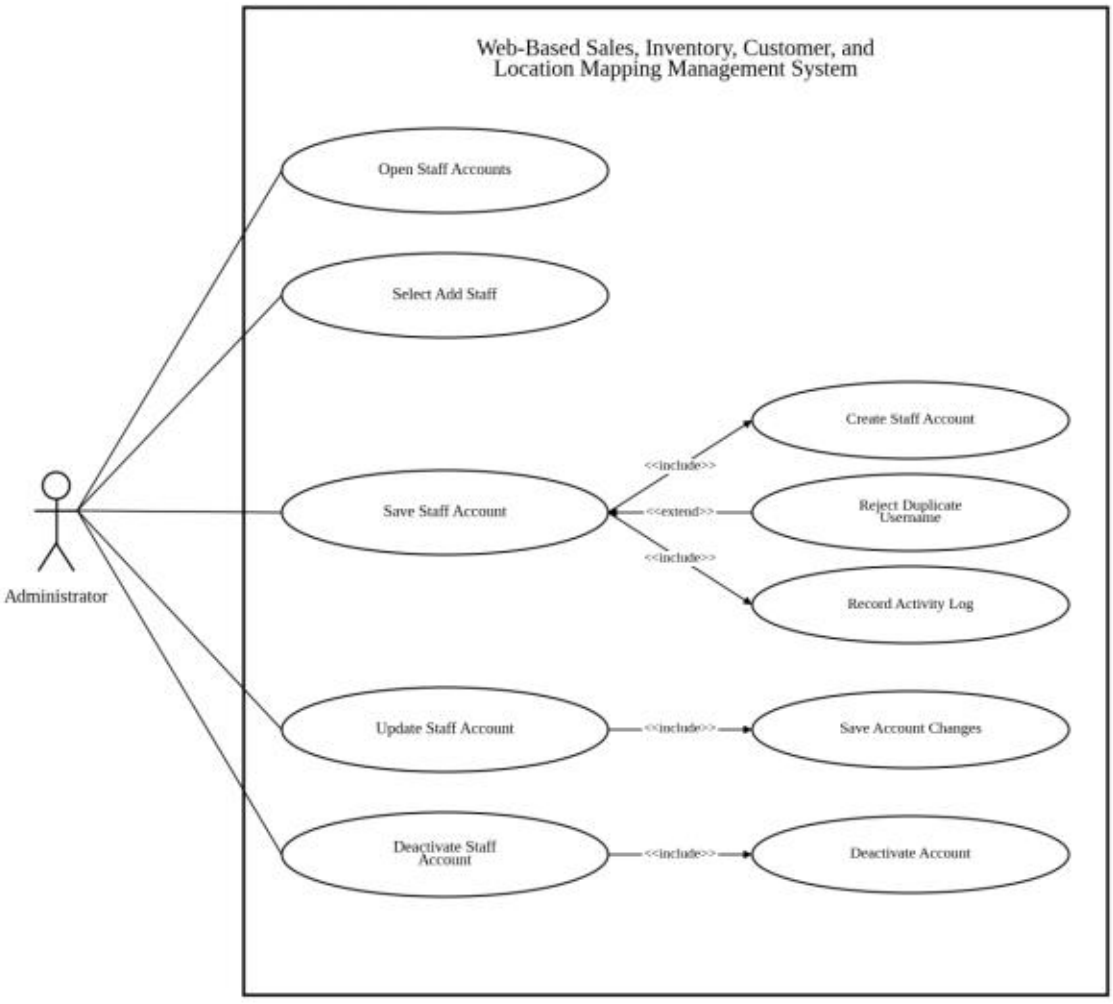


Figure 4 shows the use case diagram of Manage Staff Accounts, which is performed by: Administrator. Each step of the Normal Flow in Table 4 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 4:

Manage Staff Accounts Use Case Desription

Use Case Title:	Manage Staff Accounts
Description:	Allows the Administrator to create, update, deactivate, and manage Staff accounts and their access permissions.
Actors:	Administrator
Pre-Condition(s):	The Administrator must be logged in.
Normal Flow:	**Actor** / **System** 1. Open Staff Accounts / 1.1 Display the list of Staff accounts and the Add Staff option 2. Select Add Staff / 2.1 Display the staff account form 3. Enter the staff member's name, username, and temporary password, and save / 3.1 Validate the entries and create the account with the Staff role <<include>> Create Staff Account 3.2 <<extend>> Reject Duplicate Username 3.3 Record the action in the activity log <<include>> Record Activity Log 4. Select a Staff account to update / 4.1 Save the change and display a confirmation message <<include>> Save Account Changes 5. Select a Staff account to deactivate / 5.1 Block the account from logging in and keep the staff member's past records <<include>> Deactivate Account
Alternative Flow:	**A1. Username is already used** A1.1 The system informs the Administrator that the username already exists. A1.2 The Administrator enters a different username. **A2. Required fields are incomplete** A2.1 The system highlights the empty fields and does not save the account.

Post-Condition(s):	1. A new, updated, or deactivated Staff account is saved. 2. The action is recorded in the activity log.
Business Rules:	1. Only the Administrator can create, update, or deactivate accounts. 2. A deactivated account cannot log in, but its past records are kept. 3. A username must be unique. 4. The system has no customer accounts.

Figure 5.

Manage Products and Prices Use Case Diagram

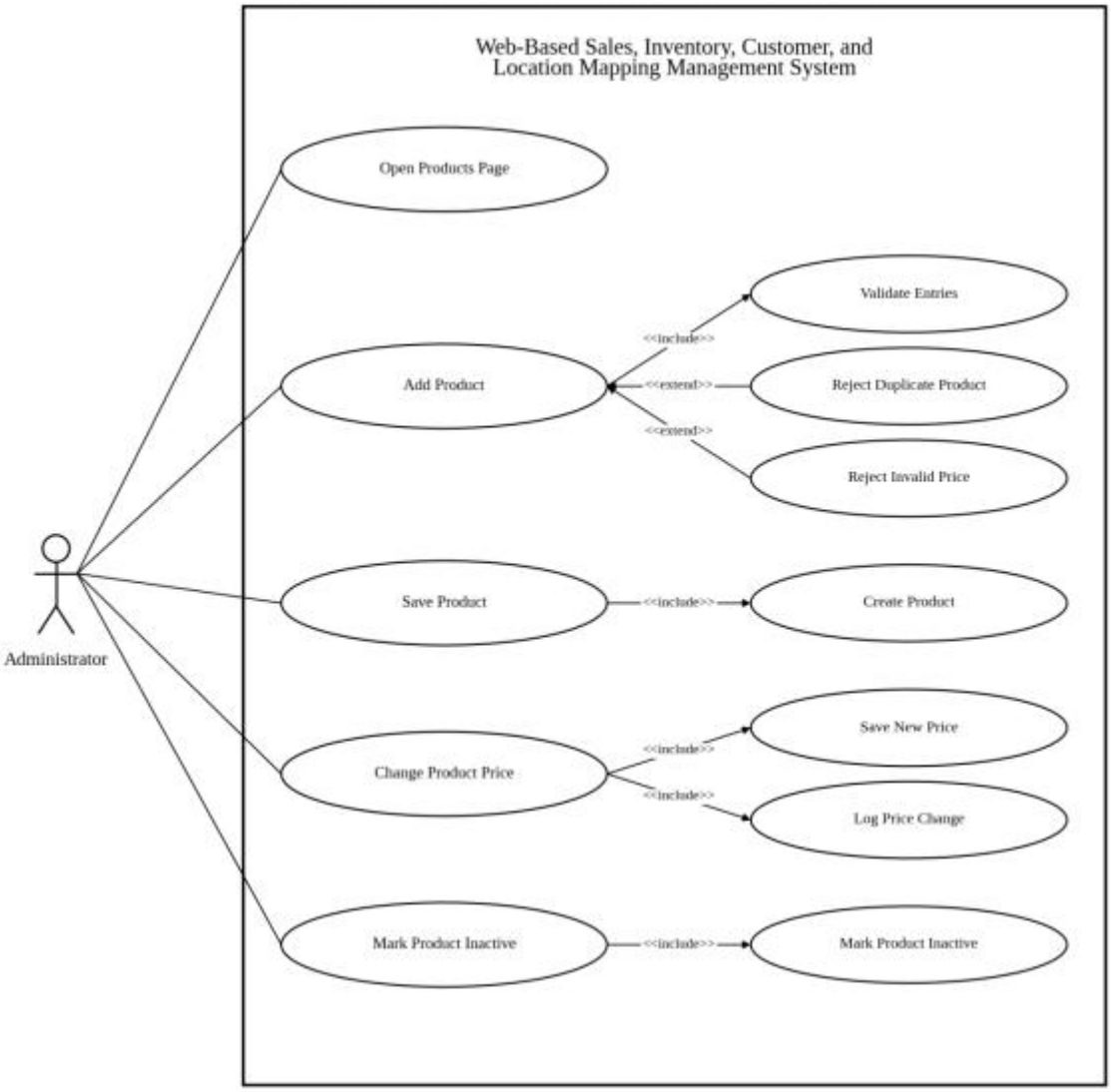


Figure 5 shows the use case diagram of Manage Products and Prices, which is performed by: Administrator. Each step of the Normal Flow in Table 5 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 5.

<table>
<tr><td colspan="2"><u>Manage Products and Prices</u></td></tr>
<tr><td>Use Case Title:</td><td>Manage Products and Prices</td></tr>
<tr><td>Description:</td><td>Allows the Administrator to maintain the product list (purified water, mineral water, containers, caps, bottles, and other supplies) and to set or update prices.</td></tr>
<tr><td>Actors:</td><td>Administrator</td></tr>
<tr><td>Pre-Condition(s):</td><td>The Administrator must be logged in.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the Products page</td><td>1.1 Display the product list and the Add Product option</td></tr>
<tr><td>2. Select Add Product and enter the product name, category, unit, and price</td><td>2.1 Validate the entries <<include>> Validate Entries
2.2 <<extend>> Reject Duplicate Product
2.3 <<extend>> Reject Invalid Price</td></tr>
<tr><td>3. Save the product</td><td>3.1 Save the product and make it available for sales, orders, and inventory <<include>> Create Product</td></tr>
<tr><td>4. Select a product and change its price</td><td>4.1 Save the new price <<include>> Save New Price
4.2 Record the change in the activity log <<include>> Log Price Change</td></tr>
<tr><td>5. Mark a product inactive when it is no longer sold</td><td>5.1 Hide the product from new sales but keep it in past transactions <<include>> Mark Product Inactive</td></tr>
</table>
</td></tr>
</table>

Alternative Flow:	**A1. Product name already exists** A1.1 The system rejects the entry and informs the Administrator. A1.2 The Administrator enters a different name. **A2. Price is invalid** A2.1 The system rejects a zero, negative, or non-numeric price. A2.2 The Administrator enters a valid price.
Post-Condition(s):	1. The product list and prices are current. 2. Past transactions keep their original prices.
Business Rules:	1. Only the Administrator can manage products and prices. 2. A product name must be unique. 3. A price must be greater than zero. 4. A price change does not alter past transactions. 5. An inactive product cannot be selected for new sales.

Figure 6.

View Dashboard Diagram

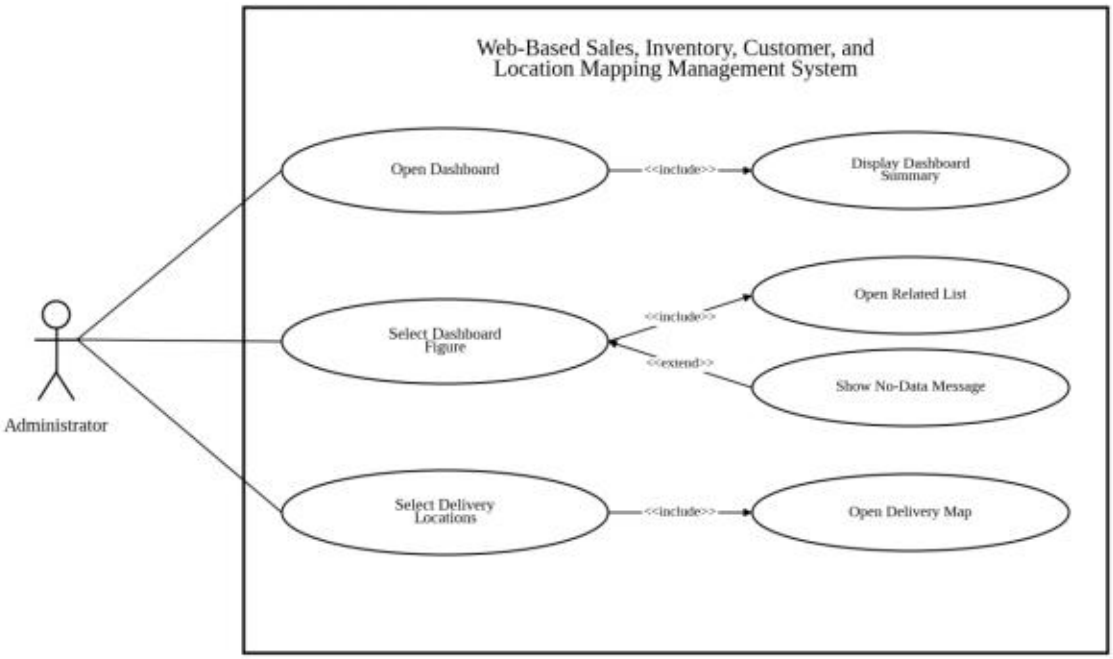


Figure 6 shows the use case diagram of View Dashboard, which is performed by: Administrator. Each step of the Normal Flow in Table 6 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 6:

View Dashboard Use Case Description

<table>
<tr><td>Use Case Title:</td><td>View Dashboard</td></tr>
<tr><td>Description:</td><td>Allows the Administrator to view a summary of total sales (including vending machine sales), pending orders, completed deliveries, available containers, low-stock products, and customer or delivery locations.</td></tr>
<tr><td>Actors:</td><td>Administrator</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. The Administrator must be logged in.
2. Transaction data exists in the system.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the dashboard</td><td>1.1 Display total sales (including vending machine sales), pending orders, completed deliveries, available containers, low-stock products, and customer or delivery locations <<include>> Display Dashboard Summary</td></tr>
<tr><td>2. Select a dashboard figure to see its details</td><td>2.1 Open the related list, such as pending orders, completed deliveries, or low-stock products <<include>> Open Related List
2.2 <<extend>> Show No-Data Message</td></tr>
<tr><td>3. Select the delivery locations figure</td><td>3.1 Open the map of customer and delivery locations <<include>> Open Delivery Map</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. No records yet
A1.1 The system displays zero values and a message that no records were found.
A2. Staff user opens the dashboard
A2.1 The system denies access because the dashboard is limited to the Administrator.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. The Administrator has current figures.
2. No transaction data is changed.</td></tr>
</table>

Business Rules:	1. Only the Administrator can view the dashboard. 2. Dashboard figures come only from recorded transactions. 3. Vending machine sales are included in the total sales. 4. The dashboard is view-only.

Figure 7.

Monitor Daily Sales and Sales Summary Use Case Diagram

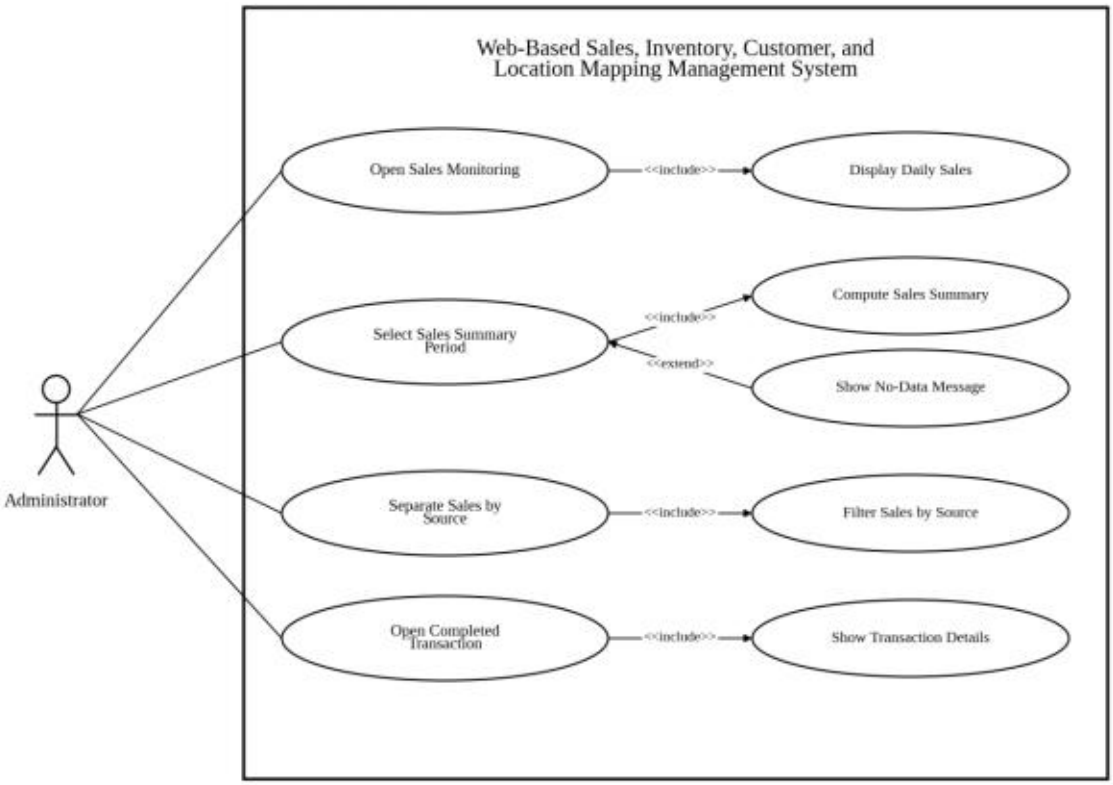


Figure 7 shows the use case diagram of Monitor Daily Sales and Sales Summary, which is performed by: Administrator. Each step of the Normal Flow in Table 7 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 7:

Monitor Daily Sales and Sales Summary Use Case Description

<table>
<tr><td>Use Case Title:</td><td>Monitor Daily Sales and Sales Summary</td></tr>
<tr><td>Description:</td><td>Allows the Administrator to monitor the daily sales of the station and its vending machines and the completed transactions, and to view daily, weekly, and monthly sales summaries.</td></tr>
<tr><td>Actors:</td><td>Administrator</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. The Administrator must be logged in.
2. Sales records exist in the system.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open Sales Monitoring</td><td>1.1 Display today’s sales from the station and the vending machines and the completed transactions <<include>> Display Daily Sales</td></tr>
<tr><td>2. Select a sales summary period (daily, weekly, or monthly)</td><td>2.1 Compute and display the sales totals for the period, with vending machine sales included <<include>> Compute Sales Summary
2.2 <<extend>> Show No-Data Message</td></tr>
<tr><td>3. Separate station sales from vending machine sales</td><td>3.1 Display the totals of each source for the same period <<include>> Filter Sales by Source</td></tr>
<tr><td>4. Open a completed transaction</td><td>4.1 Display the transaction details <<include>> Show Transaction Details</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. No records match the period
A1.1 The system displays a message that no records were found.
A2. Staff user opens Sales Monitoring
A2.1 The system denies access because it is limited to the Administrator.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. The Administrator has current daily sales and sales summaries.
2. No transaction data is changed.</td></tr>
</table>

Business Rules:	1. Only the Administrator can monitor sales. 2. Vending machine sales are included in the daily and overall sales totals and can be separated by source. 3. Sales figures come only from recorded transactions. 4. This use case is view-only.

Figure 8.

Monitor Delivery Areas Use Case Diagram

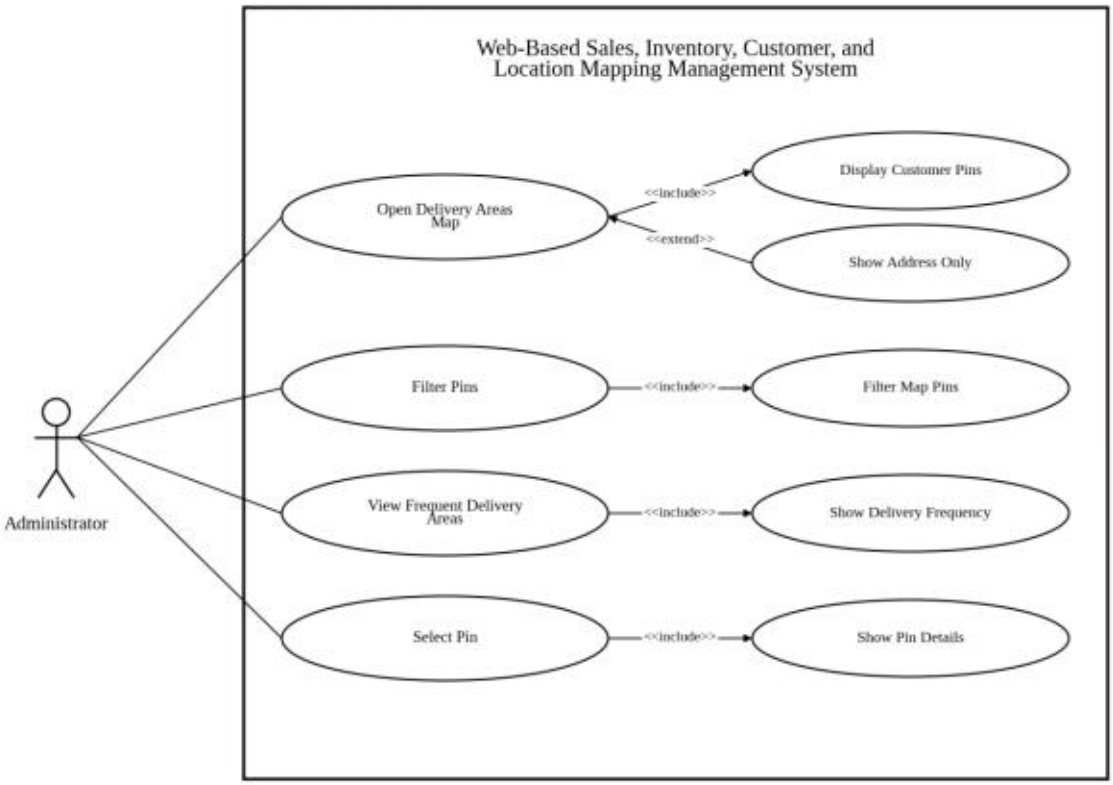


Figure 8 shows the use case diagram of Monitor Delivery Areas, which is performed by: Administrator. Each step of the Normal Flow in Table 8 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 8:

Monitor Delivery Areas Use Case Description

<table>
<tr><td>Use Case Title:</td><td>Monitor Delivery Areas</td></tr>
<tr><td>Description:</td><td>Allows the Administrator to view customer and delivery locations on a map to monitor service areas and identify areas with frequent deliveries.</td></tr>
<tr><td>Actors:</td><td>Administrator</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. The Administrator must be logged in.
2. Customer locations are saved in the system.
3. An Internet connection is available to load the map.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the Delivery Areas map</td><td>1.1 Display the station and all saved customer pins <<include>> Display Customer Pins
1.2 <<extend>> Show Address Only</td></tr>
<tr><td>2. Filter the pins by area, customer, or pending deliveries</td><td>2.1 Display the matching pins <<include>> Filter Map Pins</td></tr>
<tr><td>3. View the areas with frequent deliveries</td><td>3.1 Highlight the areas with the most deliveries <<include>> Show Delivery Frequency</td></tr>
<tr><td>4. Select a pin</td><td>4.1 Display the customer’s name, address, and open orders <<include>> Show Pin Details</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Map cannot load
A1.1 The system displays the written addresses and a message that the map is unavailable.
A2. No pin matches the filter
A2.1 The system displays a message that no locations were found.</td></tr>
<tr><td>Post-Condition(s):</td><td>The Administrator can see the service areas and the areas with frequent deliveries.</td></tr>
</table>

Business Rules:	1. The map shows saved customer locations only; the system has no live GPS tracking. 2. An Internet connection is required to load the map. 3. This use case is view-only.

Figure 9.

Generate Reports Use Case Diagram

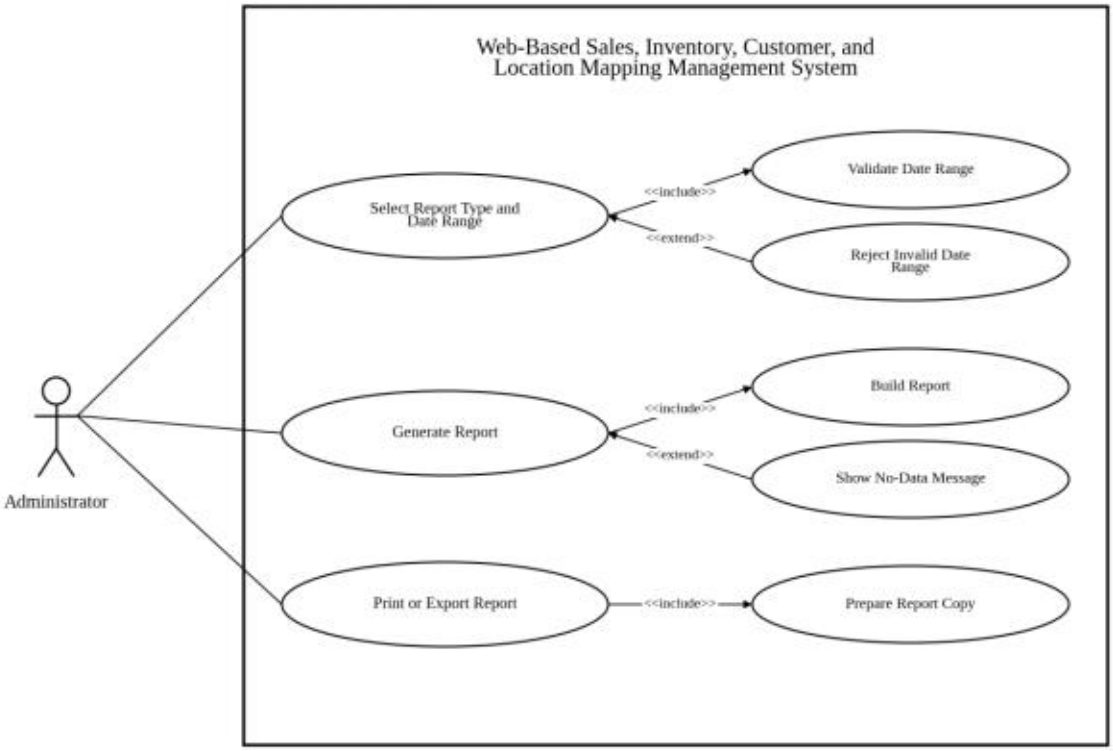


Figure 9 shows the use case diagram of Generate Reports, which is performed by: Administrator. Each step of the Normal Flow in Table 9 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 9:

Generate Reports Use Case Description

<table>
<tr><td>Use Case Title:</td><td>Generate Reports</td></tr>
<tr><td>Description:</td><td>Allows the Administrator to generate reports on sales, inventory, customers, products, payments, orders, containers, deliveries, vending machines, and locations, and to print or export them.</td></tr>
<tr><td>Actors:</td><td>Administrator</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. The Administrator must be logged in.
2. Transaction data exists in the system.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the Reports page and select a report type and a date range</td><td>1.1 Validate the date range <<include>> Validate Date Range
1.2 <<extend>> Reject Invalid Date Range</td></tr>
<tr><td>2. Generate the report</td><td>2.1 Display the report on the screen <<include>> Build Report
2.2 <<extend>> Show No-Data Message</td></tr>
<tr><td>3. Print or export the report</td><td>3.1 Prepare a printable or downloadable copy of the report <<include>> Prepare Report Copy</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. No records match the period
A1.1 The system displays a message that no records were found.
A2. Date range is invalid
A2.1 The system rejects an end date that is earlier than the start date.
A2.2 The Administrator enters a valid date range.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. The report is displayed, printed, or exported.
2. No transaction data is changed.</td></tr>
</table>

Business Rules:	1. Only the Administrator can generate reports. 2. Reports are view-only and cannot be edited. 3. Reports come only from recorded transactions.

Figure 10.

View Activity Logs Use Case Diagram

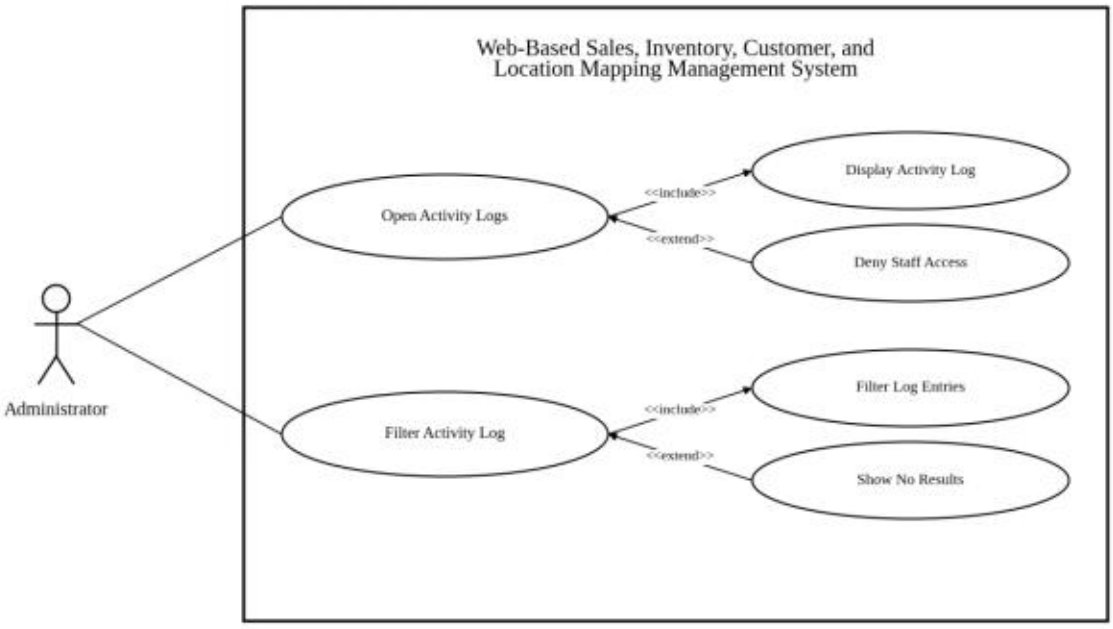


Figure 10 shows the use case diagram of View Activity Logs, which is performed by: Administrator. Each step of the Normal Flow in Table 10 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 10:

View Activity Logs Use Case Description

Use Case Title:	View Activity Logs
Description:	Allows the Administrator to monitor the important actions performed by the Administrator and Staff for accountability and security.
Actors:	Administrator
Pre-Condition(s):	1. The Administrator must be logged in. 2. Activity records exist in the system.
Normal Flow:	**Actor** / **System** 1. Open Activity Logs / 1.1 List the logins, account changes, price changes, voids, and stock adjustments with the user, action, date, and time <<include>> Display Activity Log 1.2 <<extend>> Deny Staff Access 2. Filter the log by user, action type, or date / 2.1 Display the matching log entries <<include>> Filter Log Entries 2.2 <<extend>> Show No Results
Alternative Flow:	**A1. A Staff user tries to open the activity log** A1.1 The system denies access because the log is limited to the Administrator. **A2. No log entry matches the filter** A2.1 The system displays a no-results message and suggests changing the filters.
Post-Condition(s):	The Administrator can trace important user actions.
Business Rules:	1. Only the Administrator can view the activity log. 2. Activity log entries cannot be edited or deleted.

Figure 11.

Monitor Inventory and Low-Stock Items Use Case Diagram

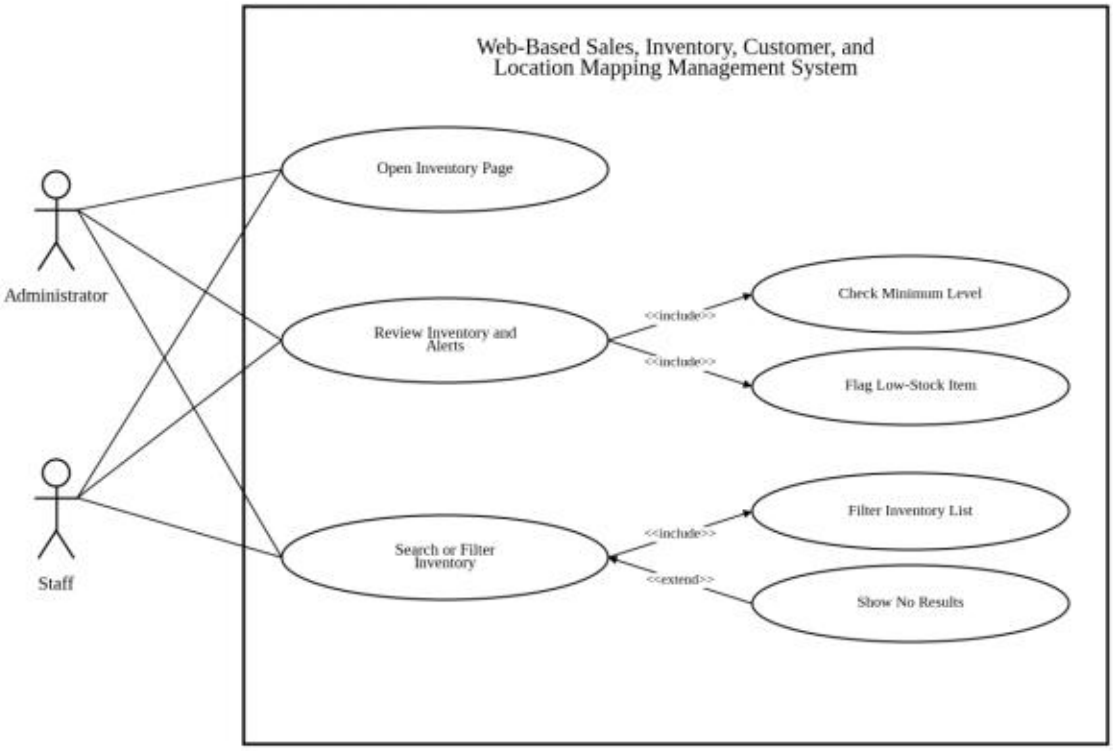


Figure 11 shows the use case diagram of Monitor Inventory and Low-Stock Items, which is performed by: Administrator, Staff. Each step of the Normal Flow in Table 11 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 11:

Monitor Inventory and Low-Stock Items Use Case Description

<table>
<tr><td>Use Case Title:</td><td>Monitor Inventory and Low-Stock Items</td></tr>
<tr><td>Description:</td><td>Allows Staff and the Administrator to monitor the quantity and availability of water containers, caps, bottles, and other station supplies, including vending machine stock, and the items that have reached their minimum level.</td></tr>
<tr><td>Actors:</td><td>Administrator, Staff</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. The user must be logged in.
2. Products exist and each has a minimum level.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the Inventory page</td><td>1.1 Display the quantity on hand of each item</td></tr>
<tr><td>2. Review the inventory list and the low-stock alerts</td><td>2.1 Compare each quantity on hand with the item’s minimum level <<include>> Check Minimum Level
2.2 Flag items at or below the minimum on the inventory page and the dashboard <<include>> Flag Low-Stock Item</td></tr>
<tr><td>3. Search or filter the inventory list</td><td>3.1 Display the matching items <<include>> Filter Inventory List
3.2 <<extend>> Show No Results</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Item reaches its minimum level after a sale
A1.1 The system automatically flags the item as low stock.
A2. No item matches the filter
A2.1 The system displays a no-results message.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. The user has seen the current stock levels and the low-stock alerts.
2. The Administrator can see which items need to be replenished.</td></tr>
</table>

Business Rules:	1. Every item has a minimum level. 2. Items at or below the minimum level are flagged as low stock. 3. This use case is view-only; stock changes are made through Manage Stock In and Stock Out.

Figure 12.

Register and Manage Customer Profiles Use Case Diagram

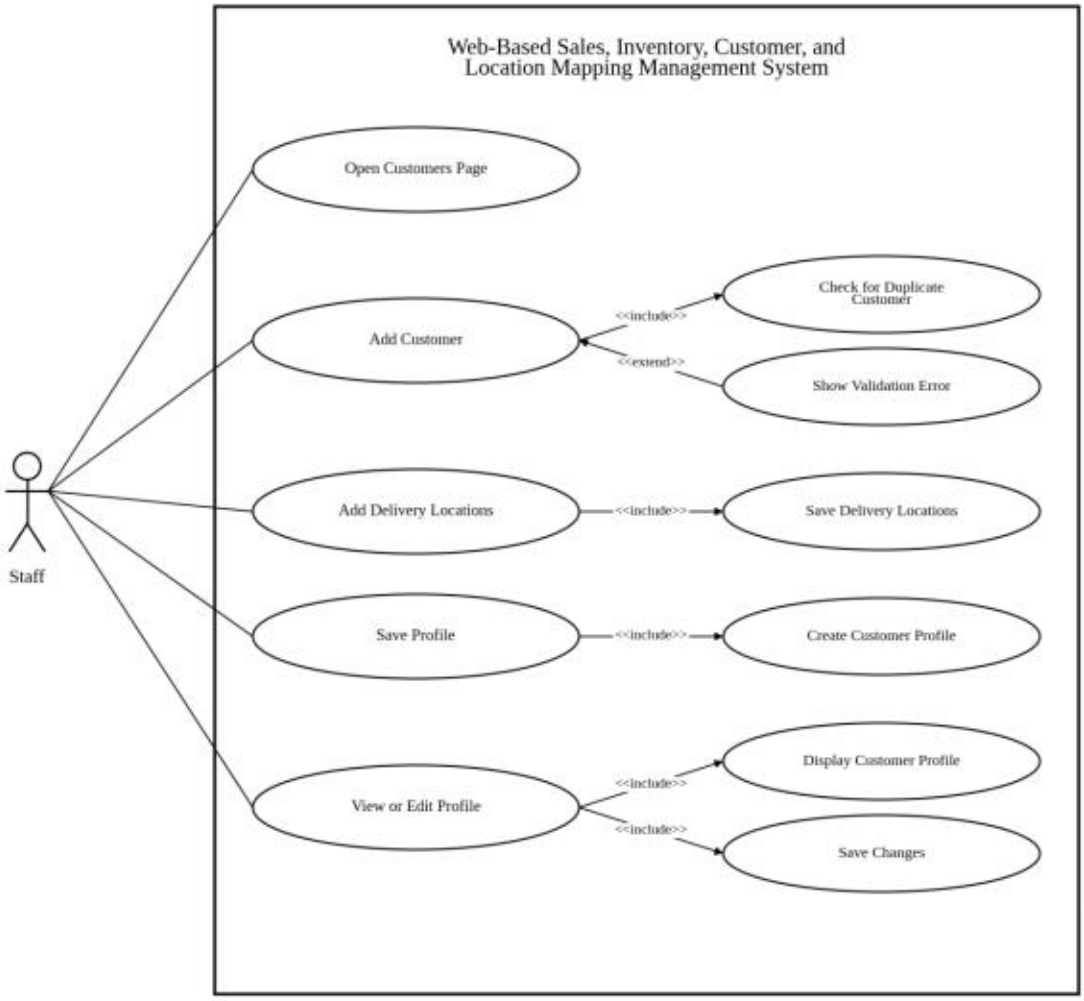


Figure 12 shows the use case diagram of Register and Manage Customer Profiles, which is performed by: Staff. Each step of the Normal Flow in Table 12 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 12:

Register and Manage Customer Profiles Use Case Description

Use Case Title:	Register and Manage Customer Profiles
Description:	Allows Staff to register customers and to view and update customer information, including contact details, delivery addresses, and customer type.
Actors:	Staff
Pre-Condition(s):	Staff must be logged in.

<table>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the Customers page</td><td>1.1 Display the customer list and the Add Customer option</td></tr>
<tr><td>2. Select Add Customer and enter the name, contact number, address, and customer type</td><td>2.1 Validate the required fields and check for a similar customer <<include>> Check for Duplicate Customer
2.2 <<extend>> Show Validation Error</td></tr>
<tr><td>3. Add one or more delivery locations</td><td>3.1 Save the delivery locations with the customer profile <<include>> Save Delivery Locations</td></tr>
<tr><td>4. Save the profile</td><td>4.1 Create the profile and display it in the customer list <<include>> Create Customer Profile</td></tr>
<tr><td>5. Open a profile to view or edit it</td><td>5.1 Display the customer’s details <<include>> Display Customer Profile
5.2 Save any edited details and display a confirmation message <<include>> Save Changes</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Required fields are incomplete
A1.1 The system highlights the empty fields and does not save the profile.
A1.2 Staff completes the entries and saves again.
A2. A similar profile already exists
A2.1 The system warns that a similar profile exists.
A2.2 Staff opens the existing profile instead of creating a new one.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. The customer profile is created or updated.
2. The profile is available for sales, orders, deliveries, and the map.</td></tr>
<tr><td>Business Rules:</td><td>1. Only logged-in Staff can manage customer profiles.
2. A customer may have more than one delivery location.
3. A customer type (walk-in, regular, or reseller) must be selected.
4. Customers do not have login accounts.</td></tr>
</table>

Figure 13.

View Customer Transaction History Use Case Diagram

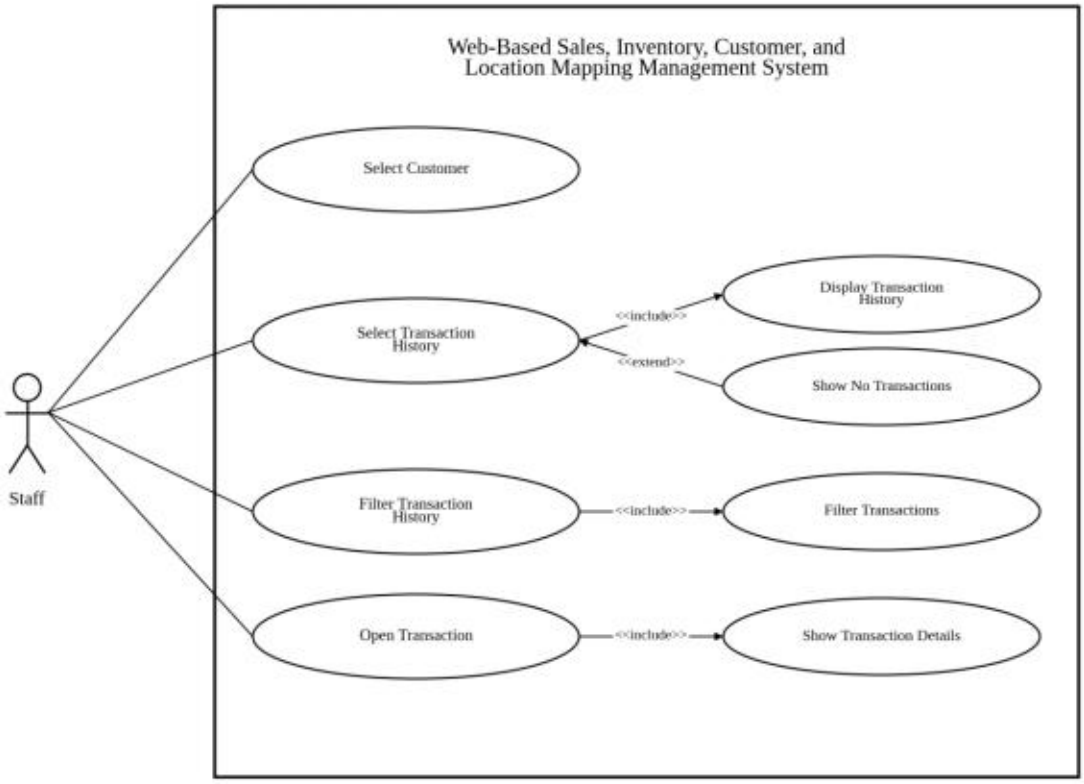


Figure 13 shows the use case diagram of View Customer Transaction History, which is performed by: Staff. Each step of the Normal Flow in Table 13 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 13:

View Customer Transaction History Use Case Description

Use Case Title:	View Customer Transaction History
Description:	Allows Staff to view the previous purchases, orders, payments, and deliveries of a customer.
Actors:	Staff
Pre-Condition(s):	1. Staff must be logged in. 2. The customer has a profile in the system.
Normal Flow:	**Actor** | **System**

	1. Open the Customers page and select a customer → 1.1 Display the customer's profile 2. Select Transaction History → 2.1 Display the customer's previous purchases, orders, payments, and deliveries <<include>> Display Transaction History 2.2 <<extend>> Show No Transactions 3. Filter the history by type or date → 3.1 Display the matching transactions <<include>> Filter Transactions 4. Open a transaction → 4.1 Display the transaction details <<include>> Show Transaction Details
Alternative Flow:	**A1. Customer has no transactions yet** A1.1 The system displays a message that no transactions were found.
Post-Condition(s):	1. Staff have seen the customer's transaction history. 2. No data is changed.
Business Rules:	1. The history comes only from recorded transactions. 2. This use case is view-only.

Figure 14.

Record Sales Transactions Use Case Diagram

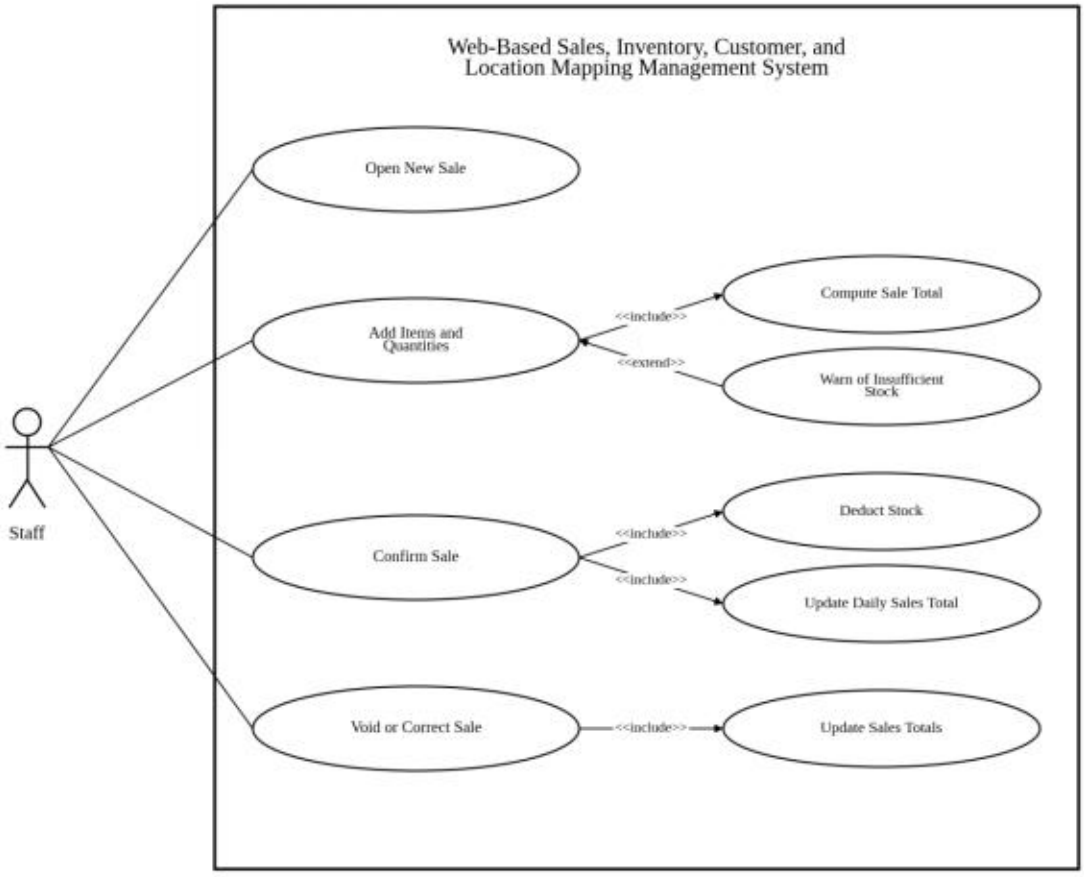


Figure 14 shows the use case diagram of Record Sales Transactions, which is performed by: Staff. Each step of the Normal Flow in Table 14 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 14:

Record Sales Transactions Use Case Description

Use Case Title:	Record Sales Transactions
Description:	Allows Staff to record purchases made at the station. Each sale is added to the daily and overall sales totals.
Actors:	Staff
Pre-Condition(s):	1. Staff must be logged in. 2. Products and prices exist in the system.
Normal Flow:	**Actor** | **System**

<table>
<tr><td></td><td>
<table>
<tr><td>1. Open New Sale and optionally select a customer</td><td>1.1 Display the sale form</td></tr>
<tr><td>2. Add the items and quantities</td><td>2.1 Compute the line totals and the grand total from the current prices <<include>> Compute Sale Total
2.2 <<extend>> Warn of Insufficient Stock</td></tr>
<tr><td>3. Confirm the sale</td><td>3.1 Save the transaction and deduct the sold supplies from inventory <<include>> Deduct Stock
3.2 Add the amount to the daily sales total <<include>> Update Daily Sales Total</td></tr>
<tr><td>4. Void or correct an entry when needed</td><td>4.1 Update the totals and record the change in the activity log <<include>> Update Sales Totals</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Insufficient stock
A1.1 The system warns that the supply item is below the requested quantity.
A1.2 Staff adjusts the quantity or removes the item.
A2. Walk-in customer
A2.1 Staff completes the sale without selecting a customer profile.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. The sale is saved.
2. Inventory is updated for the items sold.
3. The sales totals include the new amount.</td></tr>
<tr><td>Business Rules:</td><td>1. The sale total is based on the current product prices.
2. Stock cannot go below zero.
3. Voids and corrections are recorded in the activity log.
4. The system does not provide online payment.</td></tr>
</table>

Figure 15.

Record Vending Machine Sales Use Case Diagram

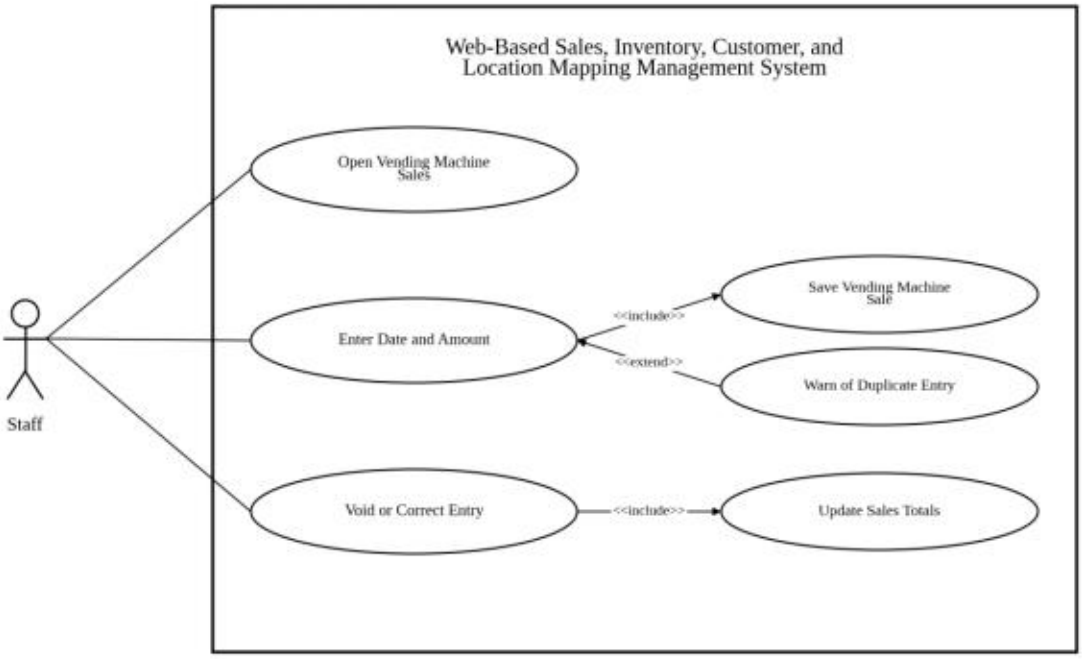


Figure 15 shows the use case diagram of Record Vending Machine Sales, which is performed by: Staff. Each step of the Normal Flow in Table 15 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 15

Record Vending Machine Sales Use Case Description

<table>
<tr><td>Use Case Title:</td><td>Record Vending Machine Sales</td></tr>
<tr><td>Description:</td><td>Allows Staff to record the sales collected from each vending machine. These are added to the daily and overall sales totals.</td></tr>
<tr><td>Actors:</td><td>Staff</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. Staff must be logged in.
2. The vending machines are registered in the system.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open Vending Machine Sales and select a machine</td><td>1.1 Display the vending machine sales form</td></tr>
</table>
</td></tr>
</table>

	2. Enter the date and the amount collected	2.1 Save the record and add it to the daily and overall sales total <<include>> Save Vending Machine Sale 2.2 <<extend>> Warn of Duplicate Entry
	3. Void or correct an entry when needed	3.1 Update the totals and record the change in the activity log <<include>> Update Sales Totals
Alternative Flow:	**A1. Duplicate vending machine entry** A1.1 The system warns that a record for the same machine and date already exists. A1.2 Staff corrects the entry or cancels it. **A2. Amount is invalid** A2.1 The system rejects a negative or non-numeric amount. A2.2 Staff enters a valid amount.	
Post-Condition(s):	1. The vending machine sales record is saved. 2. The sales totals include the new amount.	
Business Rules:	1. Vending machine sales are entered manually by Staff. 2. Voids and corrections are recorded in the activity log. 3. Vending machine sales are included in the sales totals and can be separated by source.	

Figure 16.

Manage Orders Use Case Diagram

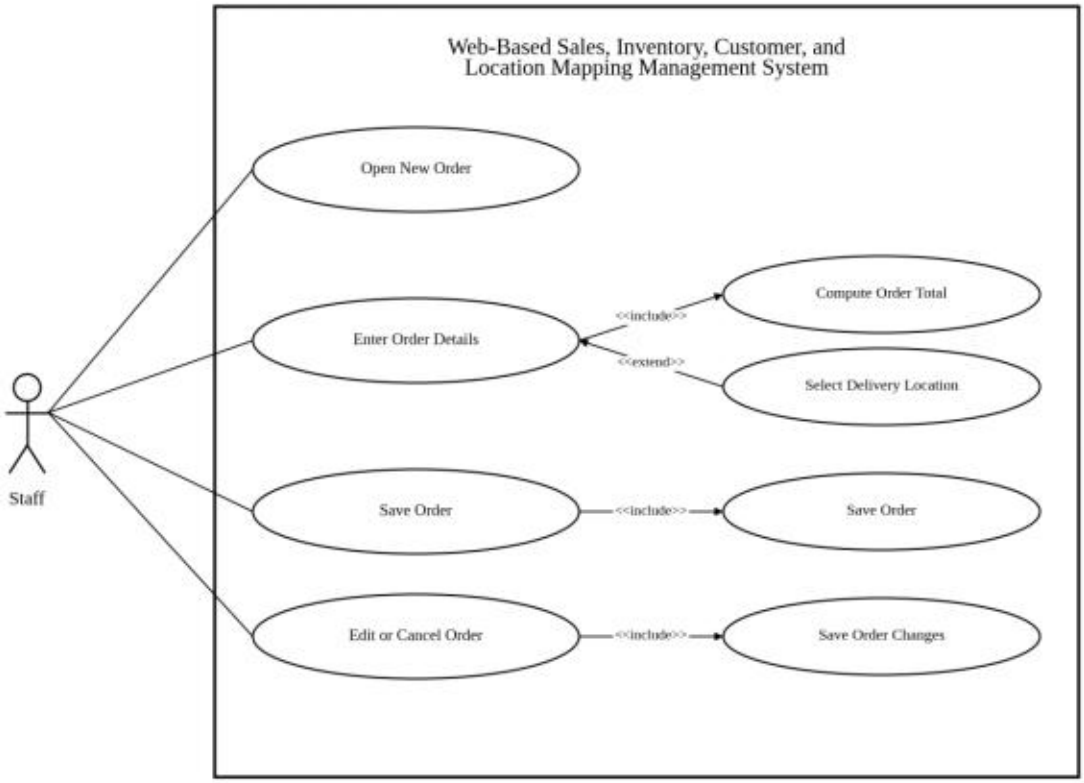


Figure 16 shows the use case diagram of Manage Orders, which is performed by: Staff. Each step of the Normal Flow in Table 16 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 16:

Manage Orders Use Case Description

Use Case Title:	Manage Orders
Description:	Allows Staff to record pickup and delivery orders with their schedules, and to edit or cancel orders when needed.
Actors:	Staff
Pre-Condition(s):	1. Staff must be logged in. 2. The customer has a profile in the system.
Normal Flow:	**Actor** | **System**

<table>
<tr><td></td><td>
<table>
<tr><td>1. Open New Order and select the customer</td><td>1.1 Display the order form with the customer’s details</td></tr>
<tr><td>2. Choose Pickup or Delivery, enter the items and quantities, and set the requested date and time</td><td>2.1 Compute the order total <<include>> Compute Order Total
2.2 <<extend>> Select Delivery Location</td></tr>
<tr><td>3. Save the order</td><td>3.1 Save the order and display it in the order list <<include>> Save Order</td></tr>
<tr><td>4. Edit or cancel the order when needed</td><td>4.1 Save the change and update any linked delivery <<include>> Save Order Changes</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Customer has no saved delivery location
A1.1 Staff adds a delivery location through Manage Customer Profiles.
A2. Order is cancelled
A2.1 Staff cancels the order and enters a reason.
A2.2 The system updates the linked delivery, if any.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. The order is saved.
2. The order is ready for preparation or delivery.
3. The customer’s transaction history is updated.</td></tr>
<tr><td>Business Rules:</td><td>1. Every order must be linked to a customer profile.
2. A delivery order requires a delivery location.
3. A cancellation requires a reason.</td></tr>
</table>

Figure 17.

Record Payments Use Case Diagram

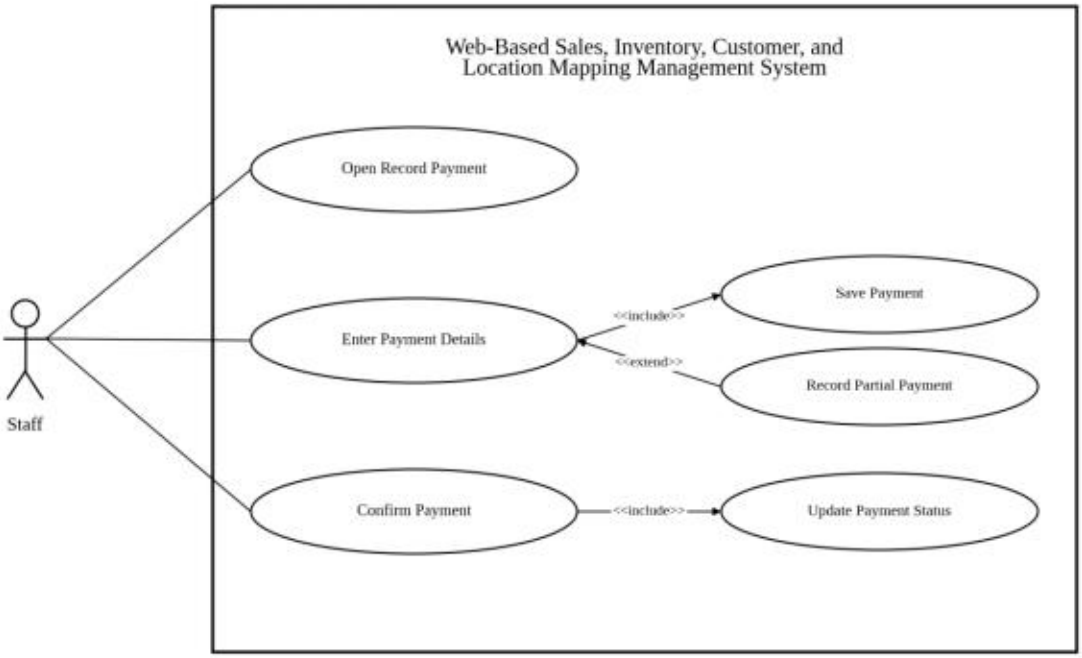


Figure 17 shows the use case diagram of Record Payments, which is performed by: Staff. Each step of the Normal Flow in Table 17 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 17:

Record Payments Use Case Description

Use Case Title:	Record Payments
Description:	Allows Staff to record the amount, method, and status of each payment of an order.
Actors:	Staff
Pre-Condition(s):	1. Staff must be logged in. 2. The order exists in the system.
Normal Flow:	**Actor** | **System** 1. Open an order and select Record Payment | 1.1 Display the payment form with the order total and the balance

	2. Enter the payment amount and method	2.1 Save the payment record <<include>> Save Payment 2.2 <<extend>> Record Partial Payment
	3. Confirm the payment	3.1 Update the payment status of the order (paid, partial, or unpaid) <<include>> Update Payment Status
Alternative Flow:	**A1. Partial payment** A1.1 Staff records the amount received. A1.2 The system shows the remaining balance. **A2. Payment on delivery** A2.1 The order stays unpaid until the delivery is completed. A2.2 Staff records the payment when the delivery is delivered.	
Post-Condition(s):	1. The payment record is saved. 2. The payment status of the order is updated.	
Business Rules:	1. Payment status is paid, partial, or unpaid. 2. Payments are only recorded; the system has no online payment gateway. 3. The amount paid cannot be more than the balance of the order.	

Figure 18.

Manage Containers Use Case Diagram

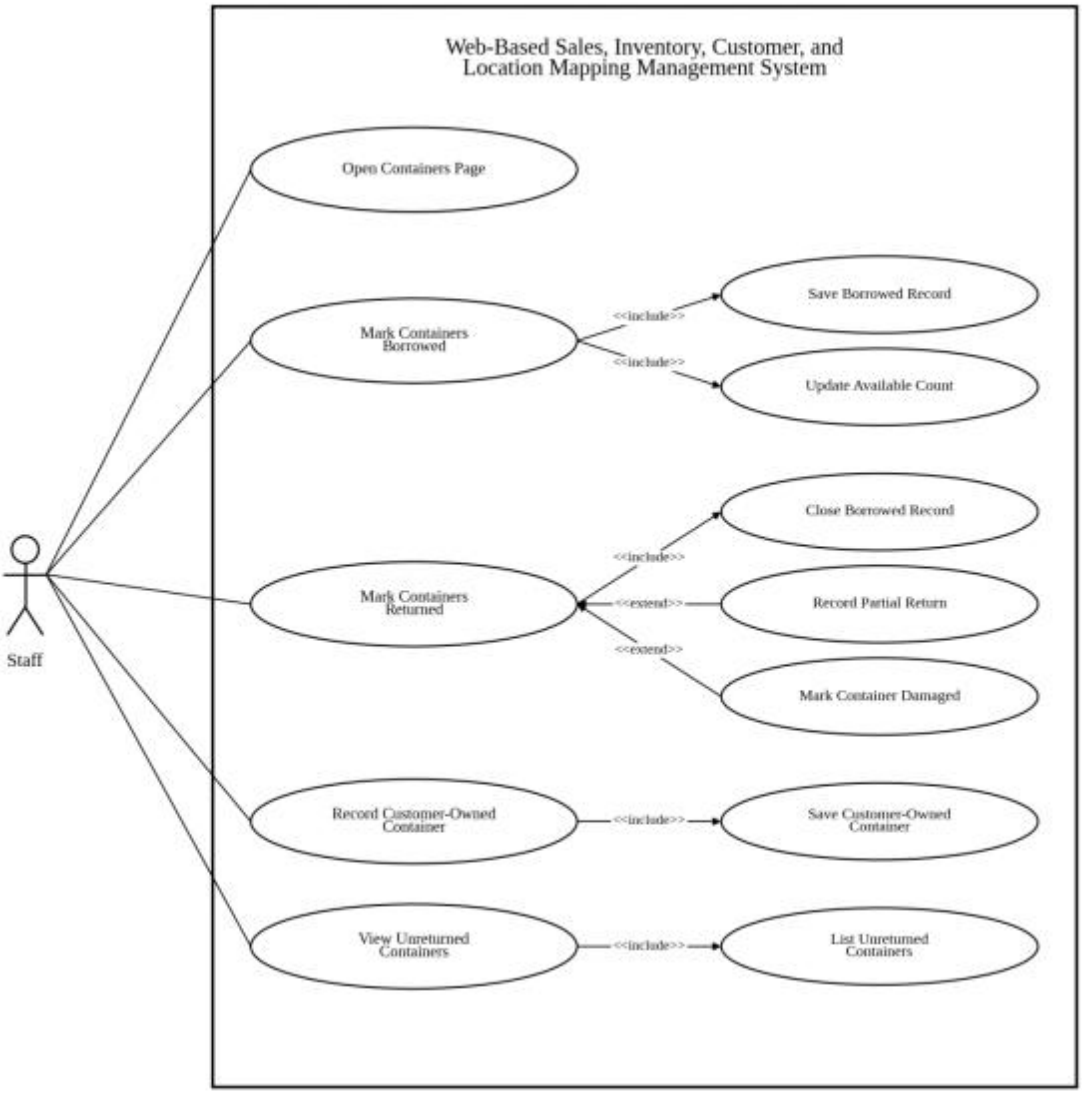


Figure 18 shows the use case diagram of Manage Containers, which is performed by: Staff. Each step of the Normal Flow in Table 18 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 18:

Manage Containers Use Case Description

Use Case Title:	Manage Containers

<table>
<tr><td>Description:</td><td>Allows Staff to track whether water containers are available, borrowed, returned, damaged, or customer-owned.</td></tr>
<tr><td>Actors:</td><td>Staff</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. Staff must be logged in.
2. Container products and customer profiles exist in the system.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the Containers page</td><td>1.1 Display the container counts and the borrowed records</td></tr>
<tr><td>2. Select the customer, container type, and quantity borrowed, and mark them as Borrowed</td><td>2.1 Save the record under the customer <<include>> Save Borrowed Record
2.2 Reduce the available count <<include>> Update Available Count</td></tr>
<tr><td>3. Select the borrowed record and mark the containers as Returned</td><td>3.1 Add the containers back to the available count and close the record <<include>> Close Borrowed Record
3.2 <<extend>> Record Partial Return
3.3 <<extend>> Mark Container Damaged</td></tr>
<tr><td>4. Record a container that the customer owns and brings for refilling</td><td>4.1 Save the record without counting the container in the station’s stock <<include>> Save Customer-Owned Container</td></tr>
<tr><td>5. View the customers who still have containers that are not returned</td><td>5.1 Display the unreturned containers of each customer <<include>> List Unreturned Containers</td></tr>
</table>
</td></tr>
</table>

Alternative Flow:	**A1. Partial return** A1.1 The customer returns only some of the containers. A1.2 The system keeps the remaining quantity as borrowed. **A2. Container is damaged** A2.1 Staff marks the returned container as Damaged. A2.2 The system removes it from the available count and keeps it in the damaged list. **A3. Not enough containers are available** A3.1 The system informs Staff that the requested quantity is more than the available count.
Post-Condition(s):	1. Container counts and per-customer container records match the actual movement of containers. 2. Damaged containers are not counted as available.
Business Rules:	1. A borrowed container must be linked to a customer. 2. A customer-owned container is not counted in the station's stock. 3. Damaged containers are listed separately. 4. A borrowed record is closed only when all its containers are returned.

Figure 19.

Record Stock-In and Stock-Out Use Case Diagram

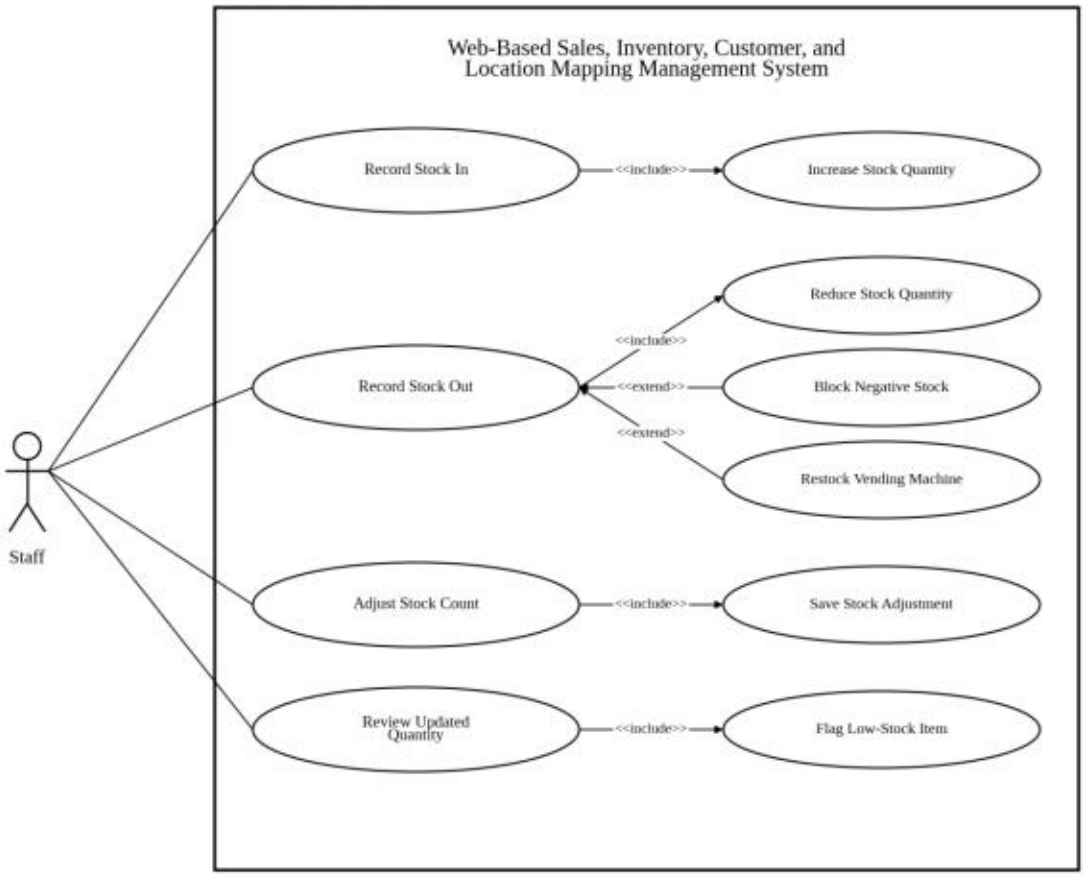


Figure 19 shows the use case diagram of Record Stock-In and Stock-Out, which is performed by: Staff. Each step of the Normal Flow in Table 19 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 19:

Record Stock-In and Stock-Out Use Case Description

Use Case Title:	Record Stock-In and Stock-Out
Description:	Allows Staff to record newly purchased or added products and supplies (stock-in) and products and supplies released, sold, consumed, or damaged (stock-out), and to correct counts after a physical check. Inventory quantities are updated.
Actors:	Staff

<table>
<tr><td>Pre-Condition(s):</td><td>1. Staff must be logged in.
2. Products exist and each has a minimum level.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the Inventory page and select Stock In, choose the product, and enter the quantity and date</td><td>1.1 Increase the quantity on hand <<include>> Increase Stock Quantity</td></tr>
<tr><td>2. Record Stock Out for supplies used, lost, or moved to a vending machine</td><td>2.1 Reduce the quantity on hand <<include>> Reduce Stock Quantity
2.2 <<extend>> Block Negative Stock
2.3 <<extend>> Restock Vending Machine</td></tr>
<tr><td>3. Correct a count after a physical check and enter the reason</td><td>3.1 Save the adjustment and record it in the activity log <<include>> Save Stock Adjustment</td></tr>
<tr><td>4. Review the updated quantity</td><td>4.1 Flag items at or below the minimum level <<include>> Flag Low-Stock Item</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Stock-out is larger than the quantity on hand
A1.1 The system rejects the entry and displays a message.
A1.2 Staff enters a valid quantity.
A2. Adjustment has no reason
A2.1 The system does not save the adjustment until a reason is entered.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. Stock levels are updated.
2. Low-stock items are flagged.</td></tr>
<tr><td>Business Rules:</td><td>1. The quantity on hand cannot be negative.
2. Sales automatically deduct the items sold from stock.
3. A stock adjustment requires a reason and is recorded in the activity log.</td></tr>
</table>

Figure 20.

Manage Deliveries Use Case Diagram

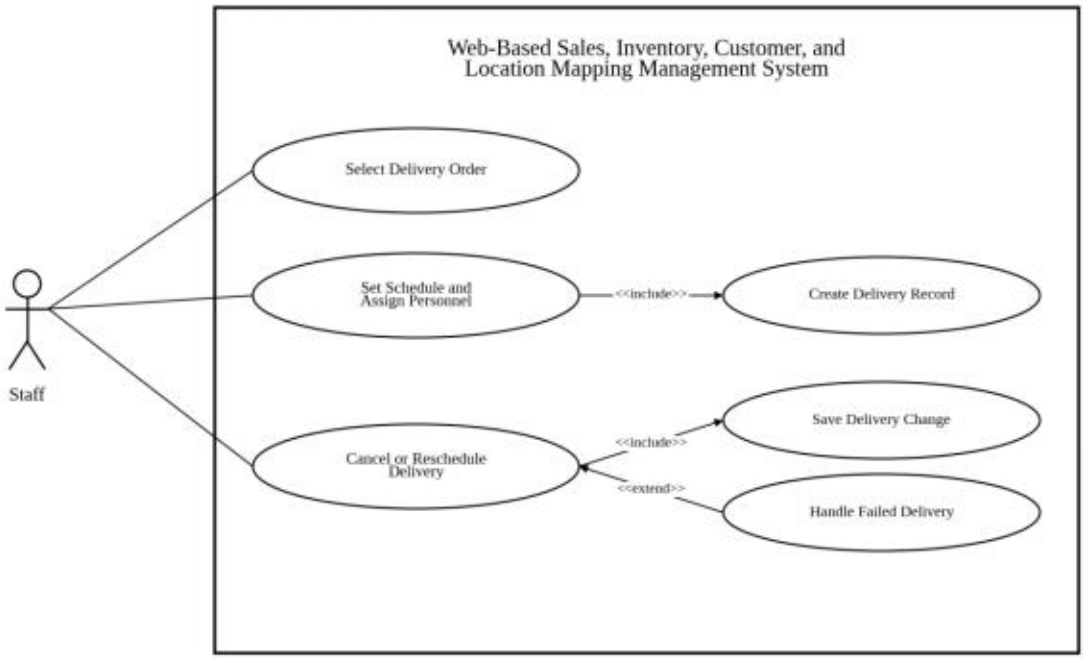


Figure 20 shows the use case diagram of Manage Deliveries, which is performed by: Staff. Each step of the Normal Flow in Table 20 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 20:

Manage Deliveries Use Case Description

Use Case Title:	Manage Deliveries
Description:	Allows Staff to create delivery records from delivery orders, assign delivery personnel, and cancel or reschedule deliveries.
Actors:	Staff
Pre-Condition(s):	1. Staff must be logged in. 2. A delivery order with a delivery location exists in the system.
Normal Flow:	**Actor** | **System**

<table>
<tr><td></td><td>
<table>
<tr><td>1. Open Deliveries and select a delivery order</td><td>1.1 Display the order details and the delivery location</td></tr>
<tr><td>2. Set the delivery schedule and assign the delivery personnel</td><td>2.1 Save the delivery with the status “Pending” <<include>> Create Delivery Record</td></tr>
<tr><td>3. Cancel or reschedule a delivery when needed</td><td>3.1 Save the change and update the order <<include>> Save Delivery Change
3.2 <<extend>> Handle Failed Delivery</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Delivery is cancelled
A1.1 Staff sets the status to Cancelled and enters a reason.
A1.2 The system updates the order.
A2. Delivery is rescheduled or reassigned
A2.1 Staff changes the date, the time, or the assigned personnel.
A2.2 The system saves the new details.
A3. Customer is not available
A3.1 Staff returns the status to Pending with a note and a new schedule.</td></tr>
<tr><td>Post-Condition(s):</td><td>1. Every delivery has a current status and an assigned person.
2. The dashboard shows the pending and completed deliveries.</td></tr>
<tr><td>Business Rules:</td><td>1. Only delivery orders can have a delivery record.
2. Delivery Personnel do not log in; Staff manage the deliveries.
3. A cancellation requires a reason.</td></tr>
</table>

Figure 21.

Track Delivery Status Use Case Diagram

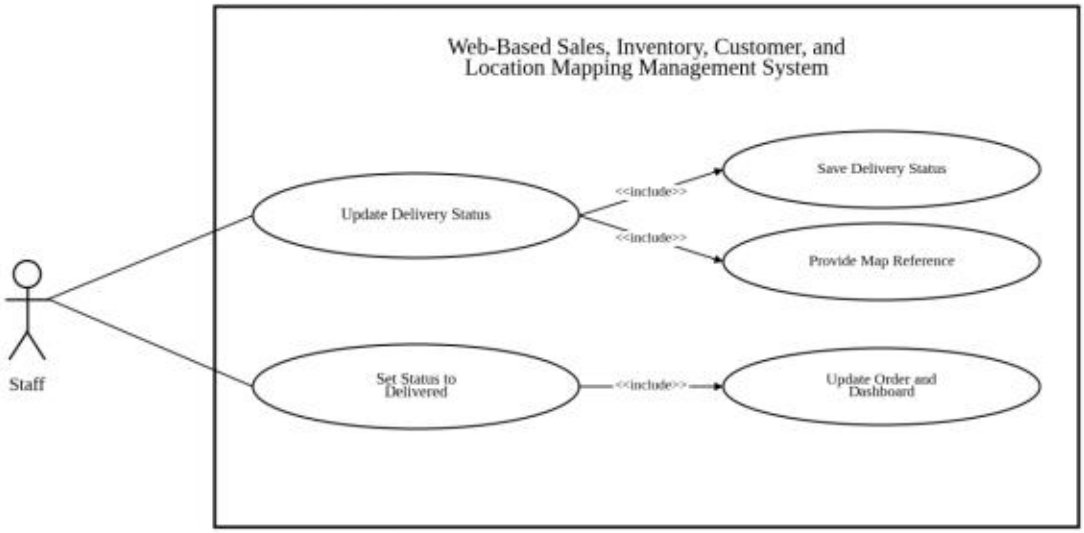


Figure 21 shows the use case diagram of Track Delivery Status, which is performed by: Staff. Each step of the Normal Flow in Table 21 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 21:

Track Delivery Status Use Case Description

Use Case Title:	Track Delivery Status
Description:	Allows Staff to track and update the status of each delivery as Pending, Preparing, Out for Delivery, Delivered, or Cancelled, and to record any payment collected.
Actors:	Staff
Pre-Condition(s):	1. Staff must be logged in. 2. A delivery record with an assigned delivery personnel exists.
Normal Flow:	**Actor** | **System**

	1. Open a delivery and update the status to Preparing and then to Out for Delivery	1.1 Save the new status <<include>> Save Delivery Status 1.2 Provide the map reference to the delivery personnel <<include>> Provide Map Reference
	2. Set the status to Delivered after the delivery personnel report the delivery, and record any payment collected	2.1 Update the order, the completed-deliveries count, and the dashboard <<include>> Update Order and Dashboard
Alternative Flow:	**A1. Delivery fails** A1.1 Staff returns the status to Pending with a note and a new schedule. **A2. Delivery is cancelled** A2.1 Staff sets the status to Cancelled and enters a reason.	
Post-Condition(s):	1. Every delivery has a current status. 2. The dashboard shows the pending and completed deliveries.	
Business Rules:	1. The delivery statuses are Pending, Preparing, Out for Delivery, Delivered, and Cancelled. 2. Delivery Personnel do not log in; Staff update the delivery status. 3. A cancellation requires a reason.	

Figure 22.

Map and Manage Customer Locations Use Case Diagram

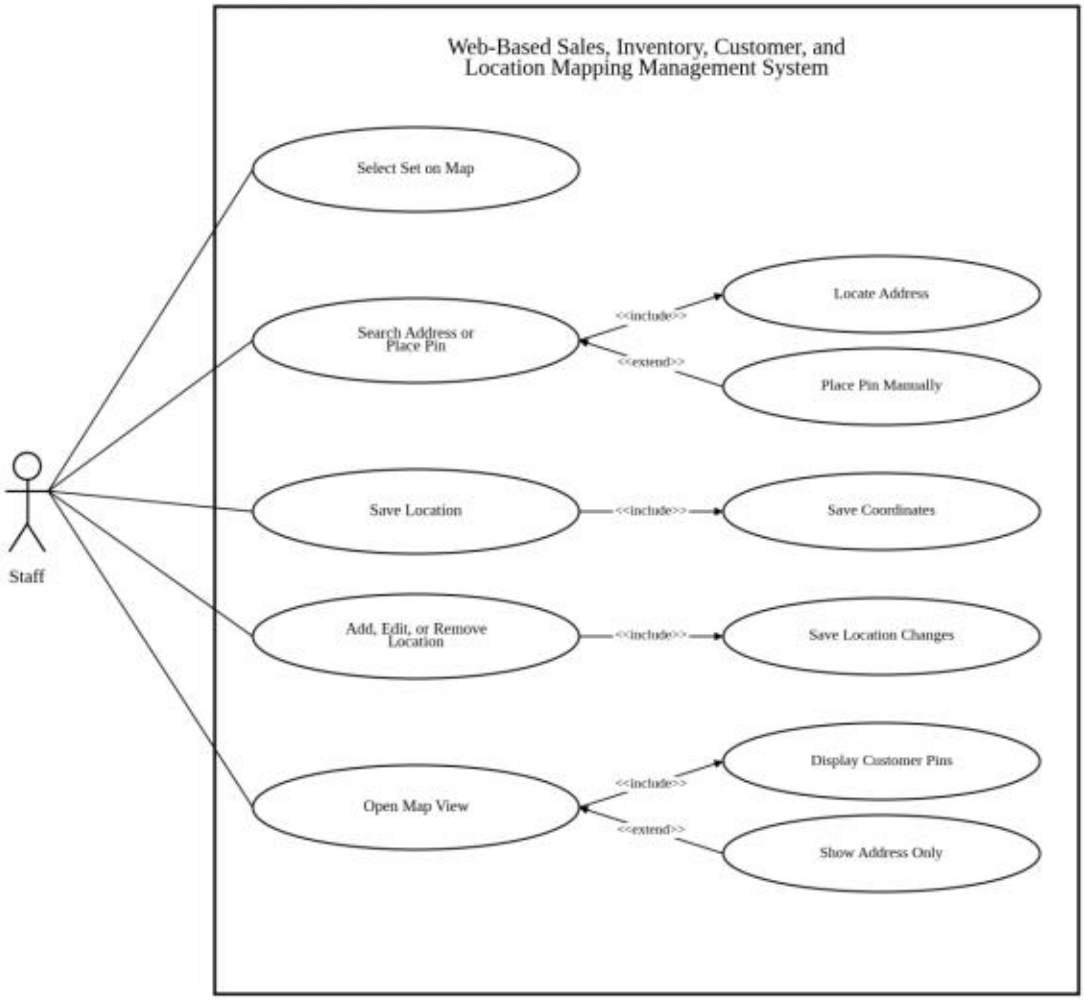


Figure 22 shows the use case diagram of Map and Manage Customer Locations, which is performed by: Staff. Each step of the Normal Flow in Table 22 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 22:

Map and Manage Customer Locations Use Case Description

<table>
<tr><td>Use Case Title:</td><td>Map and Manage Customer Locations</td></tr>
<tr><td>Description:</td><td>Allows Staff to record, locate, and visualize customer delivery addresses and the station on an interactive digital map, and to add, update, and manage customer delivery locations, including multiple addresses.</td></tr>
<tr><td>Actors:</td><td>Staff</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. Staff must be logged in.
2. The customer has a profile in the system.
3. An Internet connection is available to load the map.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open the customer’s delivery location and select Set on Map</td><td>1.1 Display the map centered on the station</td></tr>
<tr><td>2. Search for the address or place a pin at the customer’s location</td><td>2.1 Locate the address and display a pin on the map <<include>> Locate Address
2.2 <<extend>> Place Pin Manually</td></tr>
<tr><td>3. Save the location</td><td>3.1 Store the coordinates with the delivery location <<include>> Save Coordinates</td></tr>
<tr><td>4. Add another delivery location, or edit or remove an existing one</td><td>4.1 Save the change to the customer’s delivery locations <<include>> Save Location Changes</td></tr>
<tr><td>5. Open the Map view</td><td>5.1 Display the station and all saved customer pins <<include>> Display Customer Pins
5.2 <<extend>> Show Address Only</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Address is not found
A1.1 Staff places the pin manually and adds landmark notes.
A2. Pin is in the wrong position
A2.1 Staff drags the pin to the correct position and saves it.
A3. Map cannot load
A3.1 The system displays the written address and a message that the map is unavailable.</td></tr>
</table>

Post-Condition(s):	1. Customer locations are stored as coordinates and shown on the map. 2. The map reference is ready to be given to the delivery personnel.
Business Rules:	1. A customer may have more than one saved location. 2. An Internet connection is required to load the map. 3. The map shows saved customer locations only; the system has no live GPS tracking. 4. Only logged-in users can view the map.

Figure 23.

Search and Filter Records Use Case Diagram

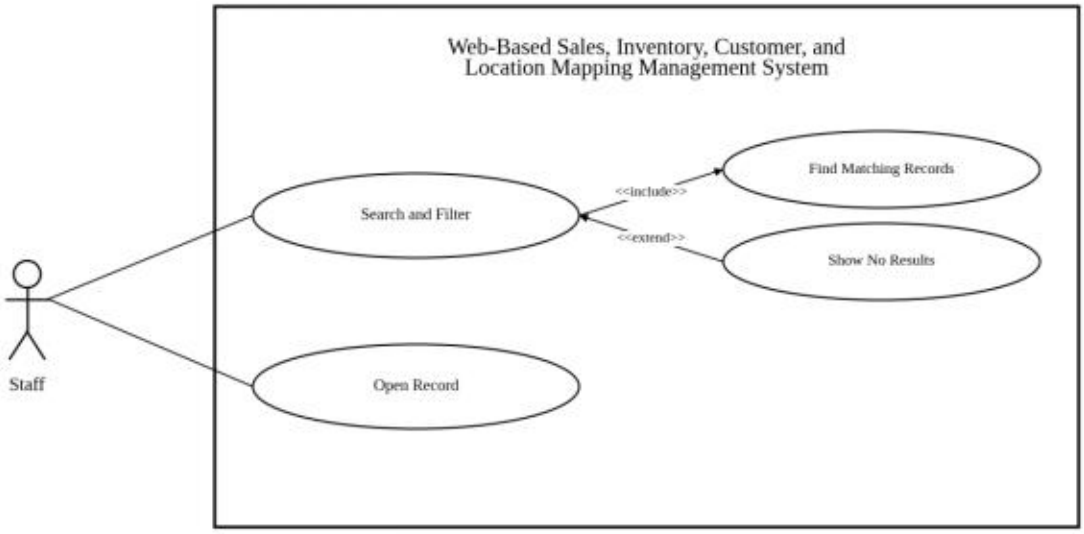


Figure 23 shows the use case diagram of Search and Filter Records, which is performed by: Staff. Each step of the Normal Flow in Table 23 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 23:

Search and Filter Records Use Case Description

Use Case Title:	Search and Filter Records
Description:	Allows Staff to search customers, products, orders, transactions, and delivery records using names, dates, status, and locations.
Actors:	Staff

Pre-Condition(s):	1. The user must be logged in. 2. Records exist in the system.
Normal Flow:	**Actor** / **System** 1. Enter a keyword in the search box and apply filters such as date, status, or customer type / 1.1 Return the matching records <<include>> Find Matching Records 1.2 <<extend>> Show No Results 2. Open a record from the results / 2.1 Display the record details
Alternative Flow:	**A1. No record matches the search** A1.1 The system displays a no-results message and suggests changing the filters.
Post-Condition(s):	The requested records are found and displayed.
Business Rules:	1. Search is view-only and does not change any record. 2. Only logged-in Staff can search records.

Figure 24.

Print Transaction Records Use Case Diagram

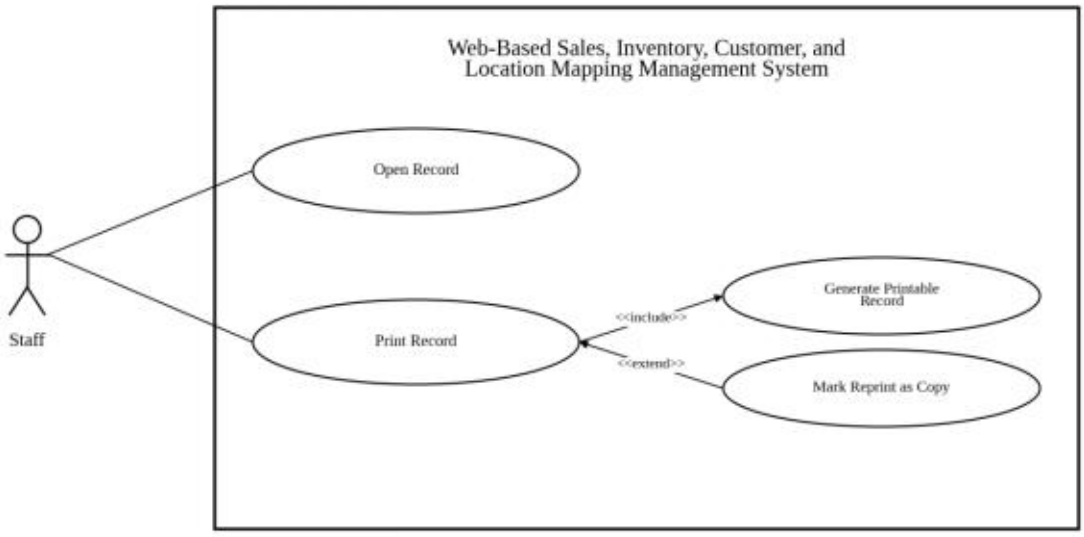


Figure 24 shows the use case diagram of Print Transaction Records, which is performed by: Staff. Each step of the Normal Flow in Table 24 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 24:

Print Transaction Records Use Case Description

<table>
<tr><td>Use Case Title:</td><td>Print Transaction Records</td></tr>
<tr><td>Description:</td><td>Allows Staff to print receipts, order records, payment records, or other transaction records when needed.</td></tr>
<tr><td>Actors:</td><td>Staff</td></tr>
<tr><td>Pre-Condition(s):</td><td>1. Staff must be logged in.
2. The sale or order exists in the system.</td></tr>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Open a sale, an order, or a payment record</td><td>1.1 Display the record details</td></tr>
<tr><td>2. Select Print</td><td>2.1 Generate the printable receipt or transaction record <<include>> Generate Printable Record
2.2 <<extend>> Mark Reprint as Copy</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. A past receipt is reprinted
A1.1 The system prints the receipt and marks it as a copy.</td></tr>
<tr><td>Post-Condition(s):</td><td>The receipt or transaction record is printed.</td></tr>
<tr><td>Business Rules:</td><td>1. A reprinted record is marked as a copy.
2. A printed record shows the recorded transaction details only.</td></tr>
</table>

Figure 25.

View Delivery Location Reference Use Case Diagram

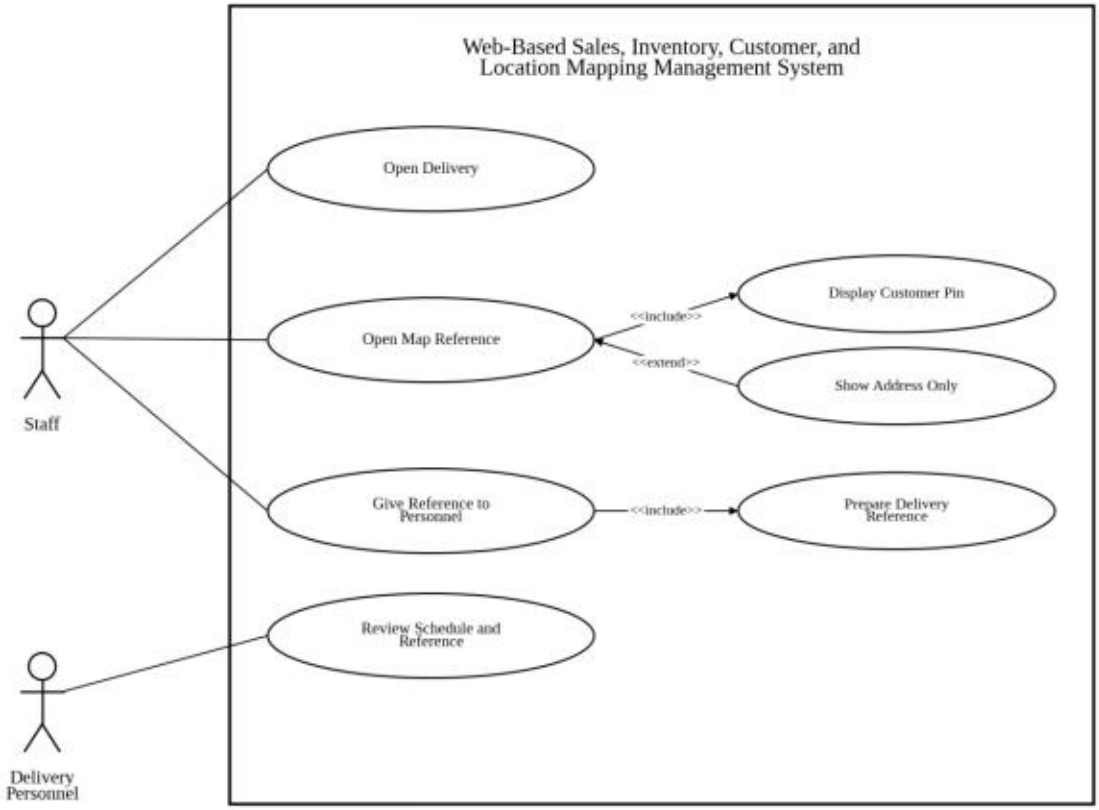


Figure 25 shows the use case diagram of View Delivery Location Reference, which is performed by: Staff, Delivery Personnel. Each step of the Normal Flow in Table 25 is shown as a use case; <<include>> marks a required sub-function and <<extend>> marks conditional handling such as an error or empty result.

Table 25:

View Delivery Location Reference Use Case Description

Use Case Title:	View Delivery Location Reference
Description:	Allows Staff to provide, and Delivery Personnel to use, a visual map reference of customer destinations and the delivery statuses, to help the delivery personnel locate delivery addresses and plan their deliveries. Delivery Personnel do not log in.
Actors:	Staff, Delivery Personnel
Pre-Condition(s):	1. A delivery with a schedule and an assigned delivery personnel exists. 2. The customer location is saved on the map.

<table>
<tr><td>Normal Flow:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. (Staff) Open the delivery</td><td>1.1 Display the schedule, the order details, the delivery location, and the delivery status</td></tr>
<tr><td>2. (Staff) Open the map reference of the delivery location</td><td>2.1 Display the station and the customer’s pin <<include>> Display Customer Pin
2.2 <<extend>> Show Address Only</td></tr>
<tr><td>3. (Staff) Give the schedule, the delivery status, and the map reference to the delivery personnel</td><td>3.1 Prepare a printable or shareable delivery schedule, delivery status, and map reference <<include>> Prepare Delivery Reference</td></tr>
<tr><td>4. (Delivery Personnel) Review the schedule, the delivery status, the address, and the map reference to locate the delivery address and plan the delivery</td><td>—</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow:</td><td>A1. Map cannot load
A1.1 The system displays the written address so that Staff can give it to the delivery personnel.
A2. Delivery is rescheduled or reassigned
A2.1 Staff gives the updated schedule to the delivery personnel.</td></tr>
<tr><td>Post-Condition(s):</td><td>The delivery personnel have the schedule, the delivery statuses, and the map reference of their deliveries.</td></tr>
<tr><td>Business Rules:</td><td>1. Delivery Personnel do not log in; they use the schedule, the delivery statuses, and the map reference that Staff provide.
2. The map shows saved locations only; the system has no live GPS tracking.
3. The delivery statuses (Pending, Preparing, Out for Delivery, Delivered, or Cancelled) are updated by Staff only.</td></tr>
</table>